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Review

Topographical design principles for osteochondral tissue engineering

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ABSTRACT

The osteochondral unit has a complex hierarchical structure where cartilage and subchondral bone show different physical, chemical, and functional properties. Replicating this structure remains a major challenge in osteochondral tissue engineering. Among various scaffold design factors, surface topography has emerged as a powerful regulator of cell behavior, but its rational integration into osteochondral constructs is still limited. This review systematically explores how topographical features affect cartilage and bone regeneration, focusing especially on feature size and anisotropy. Evidence across *in vitro* and *in vivo* studies indicates that nanoscale topographies better support chondrogenic differentiation and cartilage-like extracellular matrix formation by resembling natural cartilage, with isotropic features helping maintain chondrocyte shape. In contrast, microscale features tend to promote osteogenic differentiation, mineralization, and bone tissue organization, although osteogenic responses have also been observed on certain nanoscale topographies that mimic trabecular bone spaces. Anisotropic topographies further improve tissue-specific responses by guiding cell alignment and promoting organized matrix deposition across both cartilage and bone regions. Overall, these results highlight surface topography as a key design parameter for coordinating osteochondral regeneration. These complementary effects suggest that combining multi-scale and spatially graded topographies could imitate natural tissue architecture, enabling coordinated regeneration of cartilage and bone and offering a promising approach for functional osteochondral repair. Therefore, this review outlines practical design principles to guide the development of next-generation biomaterials for functional osteochondral repair.

1. Introduction

Advances in healthcare and improved living conditions have increased life expectancy, but this progress is accompanied by a rise in age-related musculoskeletal decline and functional impairment [1]. Degeneration of articular cartilage and weakening of bone contribute to joint disorders such as osteoarthritis (OA), which affects nearly 40% of individuals over 65 and represents a leading cause of disability worldwide, imposing a substantial socioeconomic burden [1–3]. Although OA ultimately involves multiple joint tissues, the earliest and most consequential changes occur within the osteochondral unit, a soft-to-hard tissue interface where the articular cartilage and underlying subchondral bone come together through a thin layer of calcified cartilage [4–12].

The osteochondral unit has a complex and hierarchical microscale structure essential for joint homeostasis during motor function of the musculoskeletal system [13–15]. Articular cartilage constitutes the surface layer of the joint, hence responsible for transmitting load forces to the subchondral bone [11,16]. The exceptional load-bearing efficiency and near-frictionless articulating properties of the tissue arise from depth-dependent anisotropy within the collagen fibril network. In the superficial zone, collagen fibers are oriented parallel to the articular surface, while deeper layers show progressively thicker fibers aligned perpendicularly [17]. Chondrocyte density and extracellular matrix composition also vary with depth, culminating in the mineralized calcified cartilage zone that anchors cartilage to the subchondral bone plate. Beneath this interface, the subchondral bone consists of a cortical layer and an interconnected trabecular network that distributes

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mechanical loads and maintains vascular and neural connections necessary for joint health [11,18–21].

Unlike subchondral bone, which can remodel in response to mechanical and biochemical stimuli, articular cartilage lacks vasculature and has limited intrinsic regenerative potential. As a result, injuries that disrupt the osteochondral unit, whether caused by trauma, degeneration, or surgery, often fail to heal on their own. However, due to the limited regenerative capacity of cartilage, persistent and symptomatic osteochondral injuries often require surgical intervention to restore both tissues and reestablish the native osteochondral interface (Fig. 1) [15,21,22].

Conventional management of osteochondral defects includes palliative, reparative, and restorative strategies, chosen based on lesion severity and patient factors. In advanced cases, especially those with widespread degeneration seen in osteoarthritis, more extensive procedures such as arthroplasty may be necessary [23,24]. Palliative treatments, including arthroscopic debridement, chondroplasty, and abrasion arthroplasty, mainly address mechanical irritation or inflammation and can provide temporary symptom relief. However, these methods do not restore tissue structure and usually lead to fibrocartilage formation, which lacks the mechanical and biochemical properties of native hyaline cartilage. Conversely, reparative therapies aim to stimulate natural repair or replace focal lesions through more invasive procedures like microfracture, drilling, and osteochondral autograft or allograft transplantation. While often effective for small to medium defects, these techniques are constrained by limited defect size indication, donor-site morbidity, and potential immunologic complications [25,26]. Finally, restorative therapies, such as autologous chondrocyte implantation (ACI) and its scaffold-supported variant, matrix-induced ACI (MACI), seek to regenerate hyaline-like cartilage but face persistent challenges, including restricted cell availability, chondrocyte dedifferentiation during expansion, and variable long-term outcomes [27,28].

To address these limitations, tissue engineering has emerged as a multidisciplinary approach combining biomimetic scaffolds, progenitor cells, and bioactive factors to recreate the native osteochondral environment. Scaffold-based therapies, including MACI, have shown promising regenerative potential, yet their broader clinical adoption remains limited by variability in outcomes, regulatory hurdles, and the continued preference for conservative management in early-stage disease [13,29–32].

Building on these developments, biofabrication techniques now enable the creation of interconnected scaffolds from polymers, ceramics, and metals. However, defect complexity, material limitations, and reproducibility challenges still hinder precise biomimicry of native tissues at clinically relevant scales [33,34]. These limitations are particularly pronounced in composite injuries, such as osteochondral defects [35], where monophasic scaffolds, characterized by uniform porosity, geometry, and architecture, fail to recreate the distinct biochemical and mechanical environments required for cartilage and subchondral bone regeneration [11,29].

To overcome these limitations, more advanced scaffold designs have been created. Biphasic and multiphasic structures add distinct layers to support hyaline cartilage formation, encourage vascularization, and enhance *in vivo* integration in medium and large animal models [36–39]. Similarly, continuous gradient scaffolds incorporate spatial transitions in composition, stiffness, and bioactive signaling, achieving superior structural fidelity and improved osteochondral repair compared with homogeneous designs [36,40]. These advancements have established a foundation for biologically precise scaffolds capable of mimicking zonal ECM architecture, hierarchical fiber organization, and graded mechanical properties essential for coordinated cartilage-bone regeneration [41–43].

Despite these promising developments, current multiphasic and gradient scaffolds continue to face several challenges, including incomplete integration with host tissue, mechanical displacement under physiological loading, and suboptimal orientation of the newly deposited ECM [44]. Such issues often arise from mismatched degradation or release kinetics, cellular dedifferentiation across layers, and the risk of delamination at scaffold interfaces, all of which delay reliable clinical translation [45,46].

Given these limitations, focus has shifted toward strategic biomimicry of essential structural components of native tissues, especially those that influence cell behavior via mechanotransduction. Mimicking ECM features like porosity, 3D architecture, and especially surface topography provides an effective way to direct cell adhesion, migration, and lineage determination through integrin-mediated signaling pathways [47]. Microscale topographies influence actin and stress-fiber

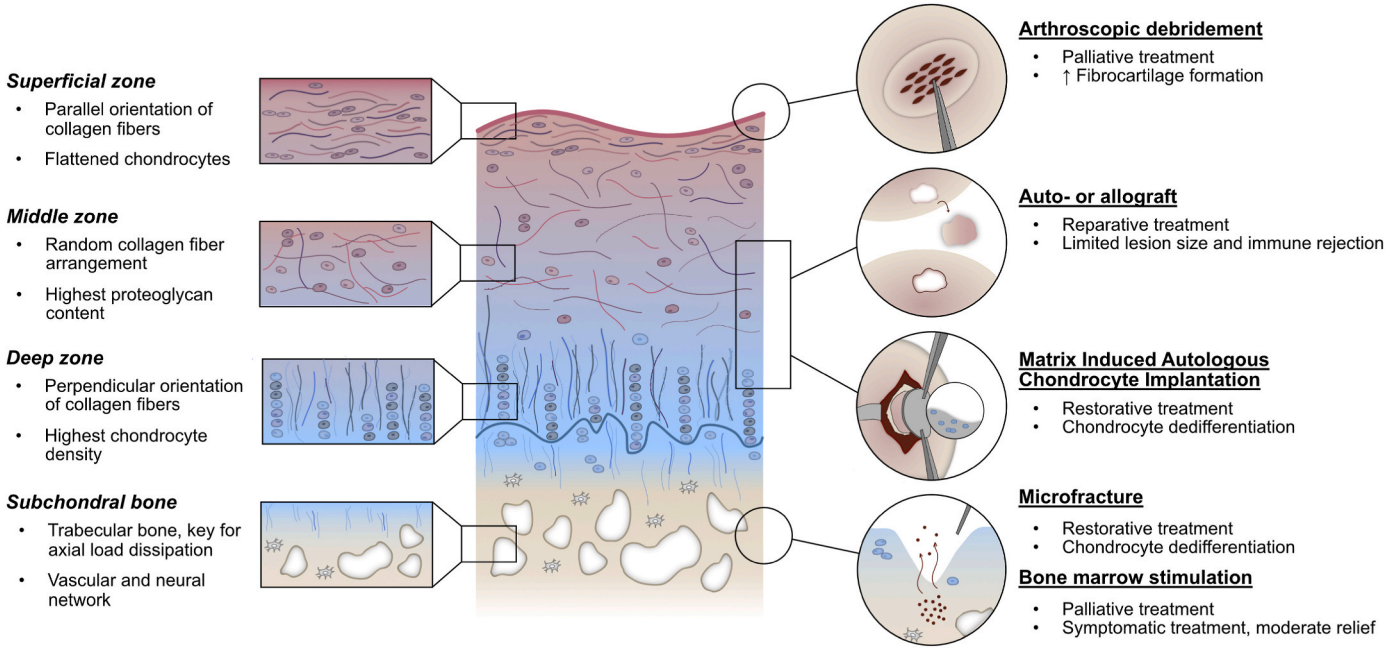


Fig. 1. Schematic representation of the targeted zones on clinical approaches for osteochondral treatment management.

organization through contact guidance, while nanoscale features impact integrin clustering and focal adhesion maturation, resulting in different downstream effects on gene expression and phenotype [48–51]. As a result, both traditional and advanced surface-engineering techniques have been used to generate controlled topographical cues, significantly enhancing cartilage and bone tissue engineering [52,53].

Despite progress in engineering complex topographical scaffolds, their clinical use still depends on overcoming significant manufacturing challenges, especially the ability to maintain pattern durability *in vivo* to ensure better cellular responses compared to unmodified versions. Retrieval analysis in preclinical models, including both small and large animals over up to 56 days, show impressive stability of the topographical features during healing around the implant, with no notable deterioration, collapse, or smoothing observed [54–56]. These findings are especially important in large animal models that closely mimic human load-bearing conditions in musculoskeletal structures.

Studies involving canine, bovine, and porcine specimens offer significant translational value because they serve as highly relevant preclinical models due to their close similarities to humans in size, weight, bone mineral density, and bone turnover rates [57]. Therefore, evaluating topographical substrates in this context can yield valuable insights into how they might perform in human osteochondral applications. However, a notable gap still exists in translating these advances into constructs that replicate the complex, hierarchical structure of the osteochondral unit.

This review, therefore, aims to provide a comprehensive analysis of topographical cues relevant to osteochondral regeneration, examining current strategies at both the nano- and micro-scale and highlighting their distinct contributions to bone and cartilage repair. Information was collected between June 2025 and January 2026 in PubMed and Scopus databases. The terms “substrate AND topography AND osteochondral AND regeneration AND scaffold” rendered 4 results. The terms “substrate AND topography AND cartilage AND regeneration AND scaffold” rendered 15 outcomes total, which were later refined by “original article” to 8 outcomes. At the same time, a similar search was conducted with the terms “substrate AND topography AND bone AND regeneration AND scaffold”, resulting in 70 outcomes total, posteriorly refined by “original article” to 58 outcomes. Additional searches including “substrate AND topography AND chondrogenic” and “substrate AND topography AND osteogenic”, rendered 41 and 397 records each, later refined by “original article” to 32 and 375 outcomes, respectively. Amongst all the outcomes combined, 170 were discarded because of duplication across databases. Exclusion criteria included studies in which non-specific topographies (e.g., general surface roughness) were addressed, lacked quantitative pattern characterization, omitted *in vitro* analyses, or involved chemical surface modifications, which eventually discarded a total of 129 outcomes. Finally, 121 publications were selected. Although the different searches were not refined by year of publication, outcomes demonstrated that topographical surfaces have gained relevance within the past 10 years, showing an exponential increase in publications since the year 2015 (Fig. 2).

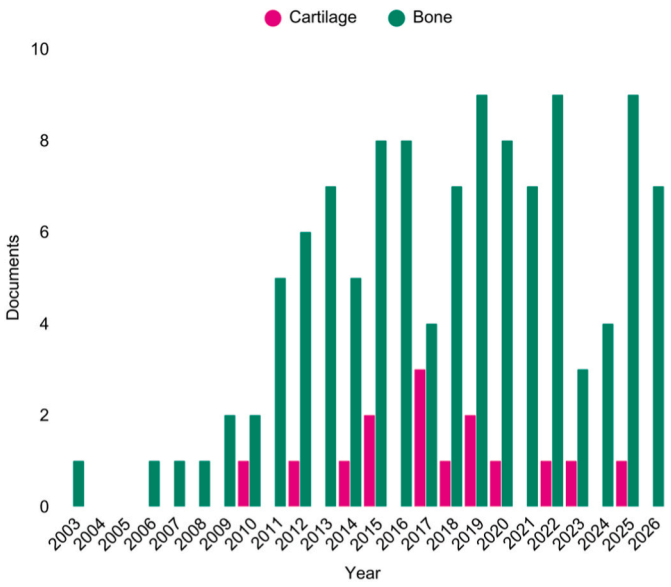


Fig. 2. Annual number of scientific publications on topographical surface design for bone and cartilage regeneration, illustrating the growing interest in biomimetic topographies and their role in osteochondral tissue engineering.

2. From structure to function: topography in tissue engineering

2.1. Topographical cues in native and engineered tissues

In native tissues, the chemical composition and structural organization of the ECM are critical in guiding tissue-specific cell-substrate interactions [58]. Among these structural cues, topography plays a central role in modulating cellular behavior. Topography refers to the physical surface structure of a material or substrate, including its patterns, textures, and elevations, typically characterized at micro- or nano-scale resolution. These features collectively define how cells perceive and interact with their surrounding environment [59].

The main source of these cues is the fibrous protein network within the ECM (e.g. collagens, proteoglycans, fibronectin) that contributes to topographical changes through differences in their diameter, pore size, or spatial arrangement (Fig. 3) [60]. For instance, collagen fibrils, which typically range from 260 to 410 nm in diameter, serve as natural nanotopographical features. These fibrils can organize longitudinally or randomly, depending on the tissue, and bundle into larger fiber networks. This bundling, together with the interstitial pores formed between the fibers, generates additional microtopographical cues [61]. Moreover, topographical cues are not limited to the ECM but they also emerge at the cellular and multicellular levels. Structures such as actin stress fibers, focal adhesion complexes, and cytoskeletal tension help define cell shape, dimensionality, and curvature, that critically influence cell response [62].

Highly organized tissues such as bone and cartilage exhibit pronounced anisotropy, which is defined as the directional variation of physical properties depending on the angle of measurement. This anisotropic organization is not only structural but also cellular, as cells themselves can adopt anisotropic and periodic arrangements, as evidenced by cytoskeletal orientation and elongated morphology on patterned substrates [63]. These cellular and extracellular arrangements collectively contribute to the topographical complexity of native tissues. Therefore, a rational and informed design strategy grounded on comprehensive tissue characterization is essential to accurately replicate these features in engineered constructs. To this end, advanced microscopy techniques have been widely used to evaluate tissue explants [64]. Scanning electron microscopy (SEM) or transmission electron microscopy provide valuable insights into the structural arrangement of the ECM components but lack the ability to quantify surface features. In contrast, atomic force microscopy, interferometry, and profilometry provide quantitative insights into surface topography [65–67]. These parameters are critical for guiding the design of biomimetic surfaces.

Inspired by the architecture of native ECM, researchers have developed engineered topographies that replicate key physical cues using techniques such as patterning and additive manufacturing (Fig. 4). Anisotropic features, including grooves and ridges, linear gratings, and aligned fibers, are typically introduced to guide cell orientation and organization. Grooves and ridges consist of parallel channels separated by elevated edges, while gratings represent repetitive linear patterns that can vary in shape, spacing, and profile [68–70]. Aligned fibers, mostly fabricated through modified electrospinning techniques, are engineered to mimic the orientation of native ECM [71]. In contrast,

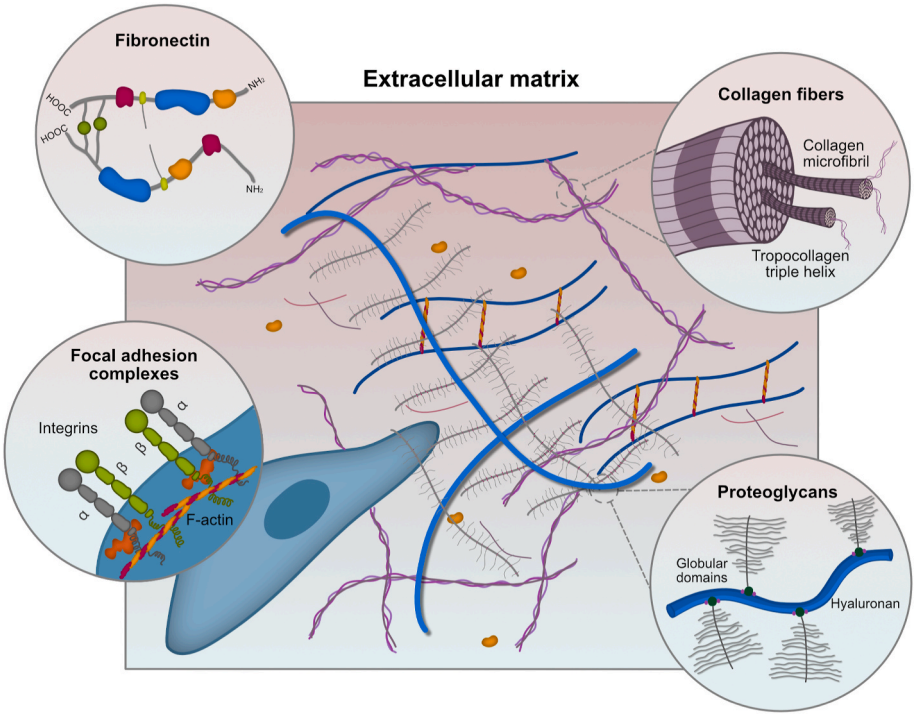


Fig. 3. Multiscale topographical cues derived from ECM and cellular structures.

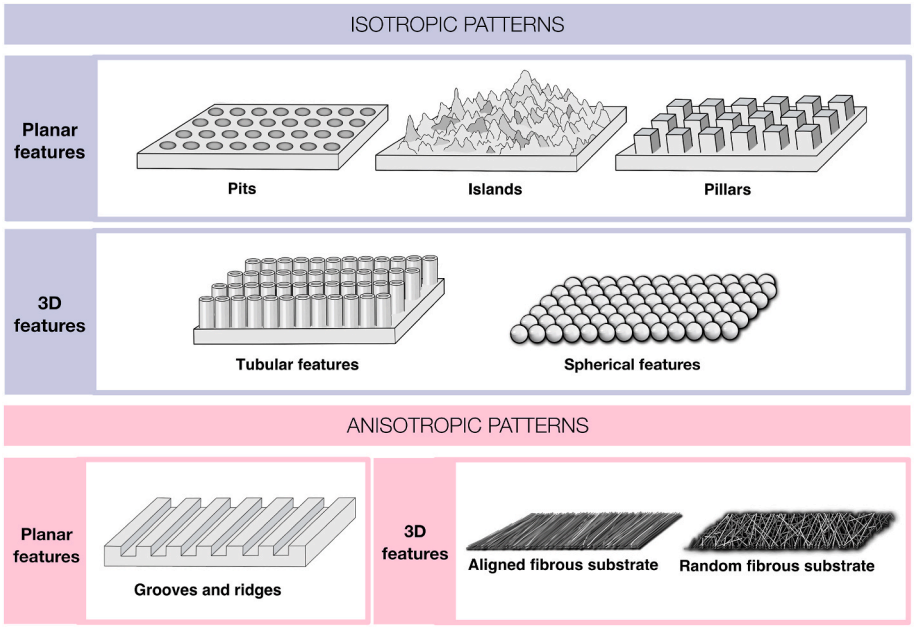


Fig. 4. Schematic summary of engineered topographies inspired by native ECM architecture.

isotropic architectures such as pits, pillars, or islands are uniform in all directions. Pits are concave indentations in a flat surface, while pillars and islands are 3D convex features projecting vertically from the substrate [72].

Given the intended application of topographically modified biomaterials as implantable medical devices, regulatory bodies have outlined standards to ensure their safety, reproducibility, and performance. The International Organization for Standardization (ISO), particularly under ISO/TS 10993, provides guidelines for the biological evaluation of medical devices, including surface characterization. These guidelines recommend both qualitative and quantitative assessment of topographical features such as roughness, smoothness, and well-defined geometries like grooves, pits, and irregular terrains. Standardized methods include SEM and profilometry, as mentioned above, along with scanning probe microscopy (SPM) and tribological analysis. Among these, SPM and profilometry offer the most precise quantification of nanoscale and microscale topographies.

2.2. *Cells respond to topographical cues*

The topographical features of the *in vivo* microenvironment are critical determinants of cellular function and tissue development. This

regulation occurs primarily through contact guidance, a process in which filopodia and membrane-bound receptors detect and respond to physical surface cues [47,51]. At the interface between the cell and its environment, integrin-ECM and cell-cell adhesion sites play a central role in detecting these cues and translating them into intracellular mechanical signals via the cytoskeleton [51].

Integrins, as heterodimeric transmembrane receptors, form the core of this mechanosensory system. Upon ligand binding, integrins undergo conformational changes that facilitate clustering and recruitment of cytoplasmic adaptor proteins, including talin, paxillin, and vinculin, to assemble focal adhesion complexes [73,74]. These dynamic structures not only serve as molecular scaffolds integrating signaling with cytoskeletal organization but also mediate mechanical communication between the cell and its environment. Specifically, they enable bidirectional force transmission, whereby cells both sense mechanical cues from the ECM and exert contractile forces back onto the substrate [74–77].

Mechanical forces detected at focal adhesions are propagated through the cytoskeleton, particularly F-actin stress fibers, and transmitted to the nucleus via the LINC complex. This process is regulated by the coordinated contributions of actin filaments, microtubules, and intermediate filaments [78–80]. Central to this process is the RhoA/ROCK signaling axis, which governs actomyosin contractility and cytoskeletal tension across focal adhesions. These mechanical signals converge on mechanosensitive pathways, including focal adhesion kinase (FAK), actin-remodeling proteins, and transcriptional co-activators such as YAP/TAZ, forming an integrated mechanotransduction network. This network controls nuclear translocation of transcription factors, chromatin remodeling, and gene expression, thereby linking extracellular topographical cues to cell behavior.

Through these pathways, topography regulates adhesion strength and organization. Microscale features that approximate or slightly exceed cell size enhance spreading by promoting integrin engagement and cytoskeletal anchoring, whereas nanoscale features near integrin-clustering dimensions are essential for focal adhesion maturation [81]. In contrast, overly dense nanotopographies may impair adhesion by promoting integrin endocytosis and destabilizing adhesion complexes [82]. Anisotropic patterns (e.g., aligned grooves or ridges) can modify cell adhesion and alignment along the imposed pattern, thereby mimicking the fibrillar organization of native ECM, whereas isotropic micropatterns often result in less organized and weaker adhesion [83].

Because adhesion strength and cytoskeletal tension are upstream of cell cycle and motility regulators, topography also modulates proliferation, morphology, and migration. For example, micro- and nanopatterned surfaces enhance proliferation in osteoblasts and MSCs, while other cell types show minimal changes under similar conditions [84,85]. Nevertheless, topographical regulation of cell shape and alignment directly influences migration, with anisotropic cues such as grooves or ridges guiding collective movement in fibroblasts and osteoblasts and enhancing both migratory velocity and directionality [86–88]. Regulation of cell shape and phenotype has also been explored in macrophages with significant effects on polarization and inflammatory response [89] although these studies have been primarily conducted in 2D [90].

Ultimately, by shaping adhesion, proliferation, and migration through integrated mechanotransduction pathways, topographical cues provide a powerful means of directing lineage-specific differentiation. As described in this review, numerous studies demonstrate that surface patterning can bias stem cell fate, positioning it as a powerful design parameter in regenerative medicine and biomaterials engineering.

3. Topographical cues for cartilage regeneration

Cartilage has emerged as a promising target for tissue-engineered regeneration due to its avascularity, aneural nature, and relatively low cellular density [91]. While its architecture is often considered less complex than that of vascularized tissues, cartilage regeneration remains challenging due to the specialized phenotype and limited proliferative capacity of chondrocytes. These cells, which constitute the sole cellular component of cartilage, exhibit a low metabolic rate, limited proliferative capacity, and a marked tendency to de-differentiate during *in vitro* expansion, that significantly hinder successful cartilage regeneration.

Since the 1990s, natural and synthetic biomaterials have been explored to develop both implantable and injectable systems in the form of fibers, meshes, and hydrogels [92–94]. However, their rapid degradation and often insufficient mechanical stability can impair cell-substrate interactions during neocartilage formation, highlighting the need for robust physical cues to support and guide tissue regeneration [95].

Chondrocytes exhibit robust mechanotransduction activity during cartilage development and homeostasis, mediated by interactions with the native ECM under physiological mechanical stimuli [96,97]. Integrins and mechanosensitive ion channels serve as key transducers of mechanical stimuli in chondrocytes, influencing cell viability, gene expression, and ECM synthesis [98]. Accordingly, leveraging the inherent mechanosensitivity of chondrocytes through engineered topographical and mechanical cues may offer a more physiologically relevant strategy for promoting chondrogenesis and facilitating cartilage repair [99].

These topographical cues can be largely classified based on their spatial organization into isotropic and anisotropic features, each mimicking different aspects of the native cartilage ECM and influencing cell behavior in distinct ways.

3.1. Isotropic patterns in cartilage regeneration

Although native articular cartilage possesses a depth-dependent anisotropy and exhibits zonal variation in ECM composition and organization, certain regions such as the proteoglycan-rich middle zone present a relatively isotropic microarchitecture, particularly in the distribution of glycosaminoglycans (GAGs) and randomly oriented collagen fibers. These uniformly organized domains contribute critically to compressive load distribution, water retention, and mechanical resilience of the tissue [99,100].

Recognition of these native isotropic features has guided the design of engineered topographies that mimic such microenvironments to direct chondrogenic behavior *in vitro*. Micro- and nano-scale isotropic features, including pits, pillars, domes, and islands, have been shown to influence chondrocyte morphology, viability, and matrix synthesis by providing spatially uniform mechanical and biochemical cues [101,102]. Additionally, curved isotropic architectures, such as spherical pores and tubular channels, have demonstrated potential to mimic the three-dimensional confinement and curvature found in native cartilage lacunae, further supporting chondrogenic differentiation of MSCs and native chondrocytes [101].

These isotropic geometries may also contribute to uniform ECM deposition, improved cell-matrix interactions, and mechanotransduction responses, particularly under compressive loading conditions [99,100,103]. Despite the dominance of anisotropic structures in native cartilage, isotropic designs offer a useful approach to reproduce uniform cell-matrix microenvironments, promoting cartilage repair by ensuring balanced mechanical stimulation and homogeneous biochemical signaling.

3.1.1. Planar isotropic features: pits, pillars, and islands

Engineered surface features such as pits, pillars, and islands are conceptually inspired by the structural complexity of cartilage ECM. While these discrete geometries are not present *in vivo*, they serve as simplified analogs to mimic the irregular and heterogeneous organization of the native matrix. In cartilage, this heterogeneity arises from variations in collagen fibril orientation, proteoglycan density, and pore

gradients across different zones [100,101]. For example, the superficial zone of articular cartilage has an effective pore size of 2-6 nm, constrained by densely packed collagen fibrils measuring 30-35 nm in diameter. With increasing depth, fibril diameter expands to 40-80 nm and spacing becomes wider, while porosity decreases to 3-5% in the calcified cartilage layer near the subchondral bone [100]. At the cellular scale, chondrocytes reside within lacunae-pericellular matrix compartments that span tens of microns in width and depth [102].

A variety of fabrication methods enable the creation of convex and concave nanoscale features on both 2D and 3D scaffolds. Conventional techniques such as polymer demixing, solvent casting, and reactive ion etching offer modest control over resolution, while advanced techniques including electron beam lithography, nanoimprinting, and templating, enable high-fidelity, reproducible patterning at the nano-to microscale [104,105].

Polymer demixing has been used to create nanopits and nanoislands with tunable dimensions by blending poly(ε-caprolactone) (PCL) and poly(methyl methacrylate) (PMMA). Depending on the blend ratio,

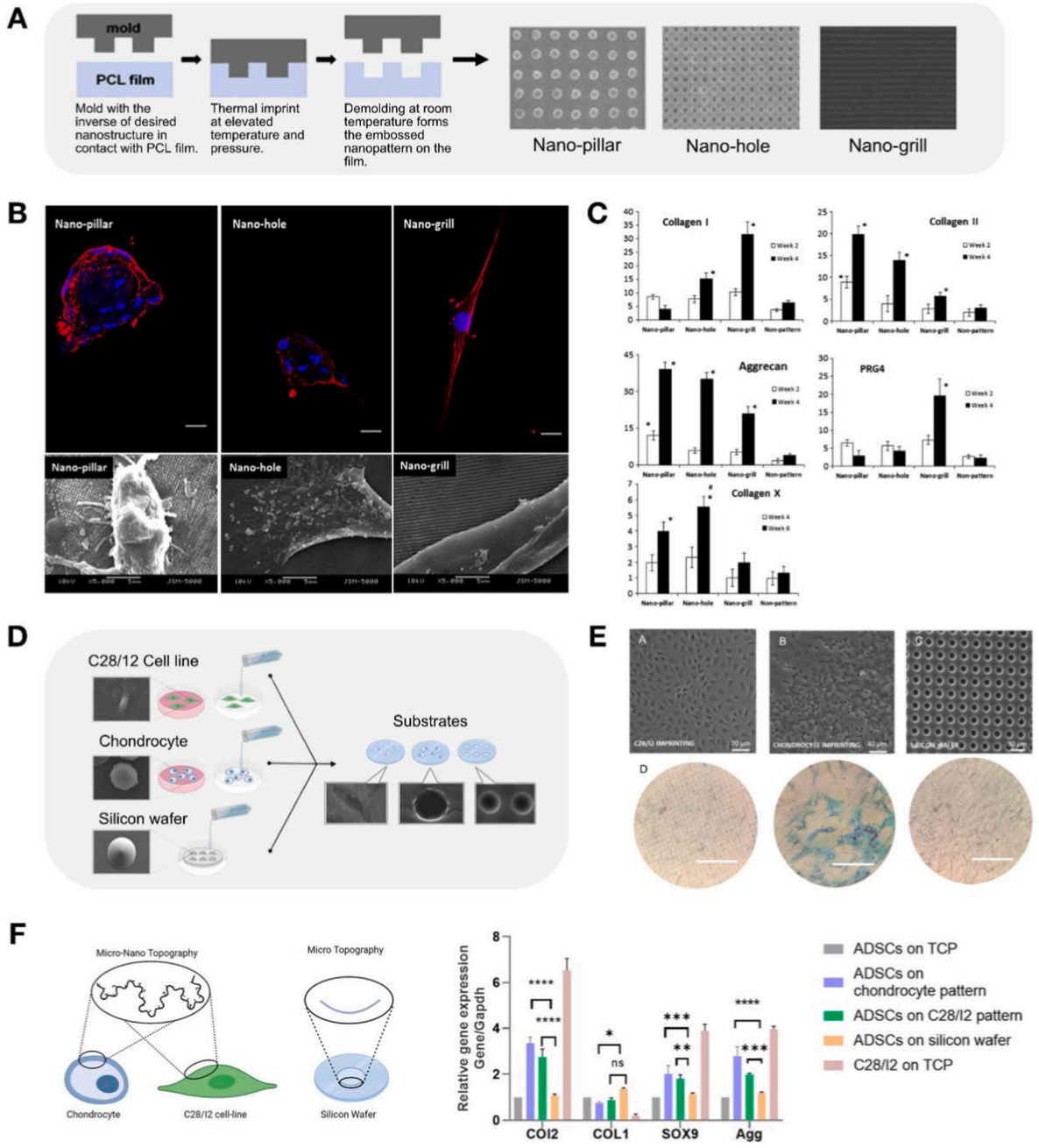


Fig. 5. Examples on isotropic surface topographies to modulate chondrogenesis and enhance cartilage regeneration. (A) Schematic representation of the thermal imprinting technique (left) and SEM micrographs (right) corresponding to nanoimprinted topographical features. (B) Confocal (top) and SEM micrographs (bottom) evidence morphological differences on cultured MSCs across patterns. On nanopillars, cells reported rounder morphology with filopodial extrusion, as shown by the cortical organization of F-actin after staining (top row, left). Scale bar: 1 mm. (C) The inherent isotropy of nanohole and nanopillar topographies allowed MSCs to undergo hyaline chondrogenic differentiation, as seen by the upregulated expression of aggrecan, collagen type II, and collagen type X [118]. Copyright 2014 Elsevier. (D) Schematic diagram of fabrication of biomimetic (C28/12) and chondrocyte imprinted substrates. (E) C28/12 and chondrocyte imprinted substrate enhance GAG deposition (bottom row). Scale bar: (a) 70 mm; (b) 40 mm; (c) 30 mm; (d) 250 mm. (F) Mimicry of chondrocyte morphology increased pro-chondrogenic gene expression [121]. Copyright 2023 IOP Publishing.

distinct surface topographies were achieved: PCL-rich blends (e.g., 75:25) formed nanopits approximately 400-700 nm wide and ~200 nm deep, while PMMA-rich blends (25:75) produced nanoislands 90-200 nm in height [106,107]. These nanostructures reduced hMSC attachment and proliferation compared to flat controls, likely due to impaired integrin clustering and focal adhesion formation when feature sizes exceed ~70 nm [108,109]. Interestingly, early expression of collagen II was detected on nanoisland surfaces by day 7, suggesting a potential for initiating chondrogenic differentiation [110].

To overcome the limitations of conventional patterning techniques, such as restricted resolution, high complexity, and limited scalability, simpler fabrication methods have been explored. For example, Foldberg et al. (2012) used a self-assembly approach to create ~248 nm pits on poly(lactic acid) (PLA) films during chloroform evaporation [111]. These surfaces upregulated SOX9 expression in adipose-derived stem cells, suggesting early chondrogenic priming [112]. However, the hydrophobicity and closed-pore structure of PLA limited cell spreading and cytoskeletal organization, reducing their overall effectiveness for cartilage-specific matrix formation. Similarly, shallow nanopits (~17 nm) have shown promise in other contexts [113]. In contrast, in murine costal chondrocytes, electron beam-lithographed nanopillars with heights of 127-190 nm promoted SOX9, COL2A1, and ACAN expression, whereas shorter pillars (62-77 nm) behaved similarly to flat controls [114]. These results identify vertical height as a dominant regulator of chondrogenic mechanotransduction by modulating cytoplasmic deformability and RhoA-dependent cytoskeletal tension, thereby governing the mechanical compliance sensed by cells [115].

Micron-scale pillars on soft polydimethylsiloxane (PDMS) [116] (1-6 μm lateral size, 2.4 μm height) enhanced chondrocyte proliferation but did not increase chondrogenic markers [117]. Thus, while microscale features may support cell viability, they require further optimization to consistently promote cartilage-specific differentiation and ECM production.

In contrast to microscale features, nanoscale topographies have demonstrated greater potential for guiding stem cell fate and mimicking the native ECM. Scalable methods such as thermal nanoimprinting have been employed to produce reproducible nanotopographies over large areas (Fig. 5A). Arrays of 225 nm nanoholes and 250 nm nanopillars with 300 nm spacing modulated MSC morphology in distinct ways: nanopillars induced a rounded morphology with prominent filopodia, whereas nanoholes produced polygonal shapes (Fig. 5B) [118]. These morphological responses align with prior observations that interfacial forces influence cytoskeletal organization and cell positioning between features [119]. Gene expression analysis confirmed upregulation of SOX9, aggrecan, and COL2A1 on nanopillars, emphasizing the importance of vertical dimensions (127-330 nm) in promoting cartilage-specific integrin signaling (Fig. 5C) [120].

3.1.2. 3D isotropic features: tubular and spherical structures

In native cartilage, ECM components present both spherical and tubular geometries. Type II collagen molecules assemble into nanoscale fibrils that bundle into tubular fibers, providing tensile strength and structural integrity [122]. Concurrently, aggrecan aggregates with hyaluronic acid to form hydrated networks containing spherical proteoglycan clusters, essential for resisting compressive loads. Chondrocytes themselves typically adopt a rounded morphology within lacunae, a feature associated with their differentiated state [123].

To replicate the tubular architecture of native cartilage ECM, titanium-based substrates have been engineered due to their biocompatibility, mechanical strength, and integration capacity [124–126]. Vertically aligned TiO_2 nanotube arrays fabricated via anodization [127] have shown promise in preserving the rounded morphology of hyaline chondrocytes, particularly at a pore size of ~70 nm. This morphology, supported by cortical actin ring formation, correlates with maintenance of the chondrogenic phenotype, compared to smooth surfaces [128]. Nanotubes with diameters between 50 and 100 nm significantly enhanced GAG secretion, aligning with increased prevalence of rounded chondrocytes and improved cartilage-specific matrix assembly [129].

Spherical microtopographies have been explored to replicate the chondrocyte niche and support cartilage-specific functions. Poly(L-lactic acid) (PLLA) substrates with uniform microspherulites (~30-50 μm) fabricated via thermal annealing and crystallization were shown to preserve the chondrogenic phenotype by sustaining collagen II and aggrecan expression while markedly suppressing collagen I, in contrast to amorphous controls that promoted rapid dedifferentiation [130]. These microtopographies reduced stress-fiber formation and promoted α5β1-mediated adhesion, a hallmark of native cartilage, thereby limiting cytoskeletal tension and delaying the α2β1-driven shift toward a fibroblastic state. This approach also enables modulation of surface stiffness and roughness, making it a promising strategy for recreating osteochondral gradients [131]. Similar-sized microfeatures have been shown to enhance collagen II deposition, though they are more commonly applied in grooved formats [132].

Cell-imprinted spherical pits have emerged as a promising isotropic topography for directing chondrogenic differentiation. Taheri et al. (2023) fabricated substrates with pit arrays mimicking the shape of native chondrocytes, either spherical or spindle-like, using photolithography and nanoimprinting (Fig. 5D) [121]. When MSCs were cultured on these surfaces, the spherical imprints significantly enhanced expression of SOX9, collagen II, and aggrecan compared to both spindle-shaped and standard lithographic controls (Fig. 5E–F). These outcomes indicate that replicating the natural cell-scale geometry, not only feature size, can more effectively preserve the chondrogenic phenotype and promote cartilage-specific matrix production.

3.2. Anisotropic patterns in cartilage regeneration

The anisotropic architecture of native articular cartilage, characterized by depth-dependent collagen alignment, has been described previously in Section 2.1. This structural organization plays a critical role in mechanical function, with collagen fibrils aligned parallel to the surface in the superficial zone to resist shear, and transitioning to oblique and perpendicular orientations in the deeper zones for anchorage into subchondral bone [133,134]. Inspired by this native anisotropy, engineered surface features such as grooves, ridges, and aligned fibers, have been developed to direct stem cell alignment, support chondrogenic differentiation, and maintain cell phenotype. These anisotropic cues enhance cartilage-specific gene expression and ECM production in a manner that is highly dependent on topographical geometry, feature size, and substrate stiffness [114,118,135].

3.2.1. 2D anisotropic features: grooves and ridges

The nanoscale alignment of type I collagen fibrils not only confers mechanical strength but also provides potent topographical cues that modulate focal adhesion dynamics and cytoskeletal organization. Mimicking this native anisotropy, ordered groove-ridge structures are widely used to replicate aspects of the collagen lattice found in load-bearing tissues. In articular cartilage tissue engineering, such topographies recapitulate the aligned architecture of the superficial zone, directing cell orientation, elongation, and lineage commitment through contact guidance. Groove widths as small as ~50 nm can affect cell behavior, although ridges narrower than ~70-100 nm may hinder integrin binding and reduce adhesion [136]. Deeper grooves generally promote greater elongation and alignment, demonstrated across multiple cell types [137–139]. Dual-scale groove patterns fabricated via femtosecond laser processing enhanced alignment and upregulated SOX9 and SPP1 compared to flat surfaces [140].

However, not all groove configurations are beneficial for cartilage regeneration. Grooves ≥317 nm on PLGA induced strong elongation but also upregulated osteogenic markers such as RUNX2 and bone sialoprotein (Fig. 6A–B) [141]. Similarly, grooves on PCL films (~250 nm

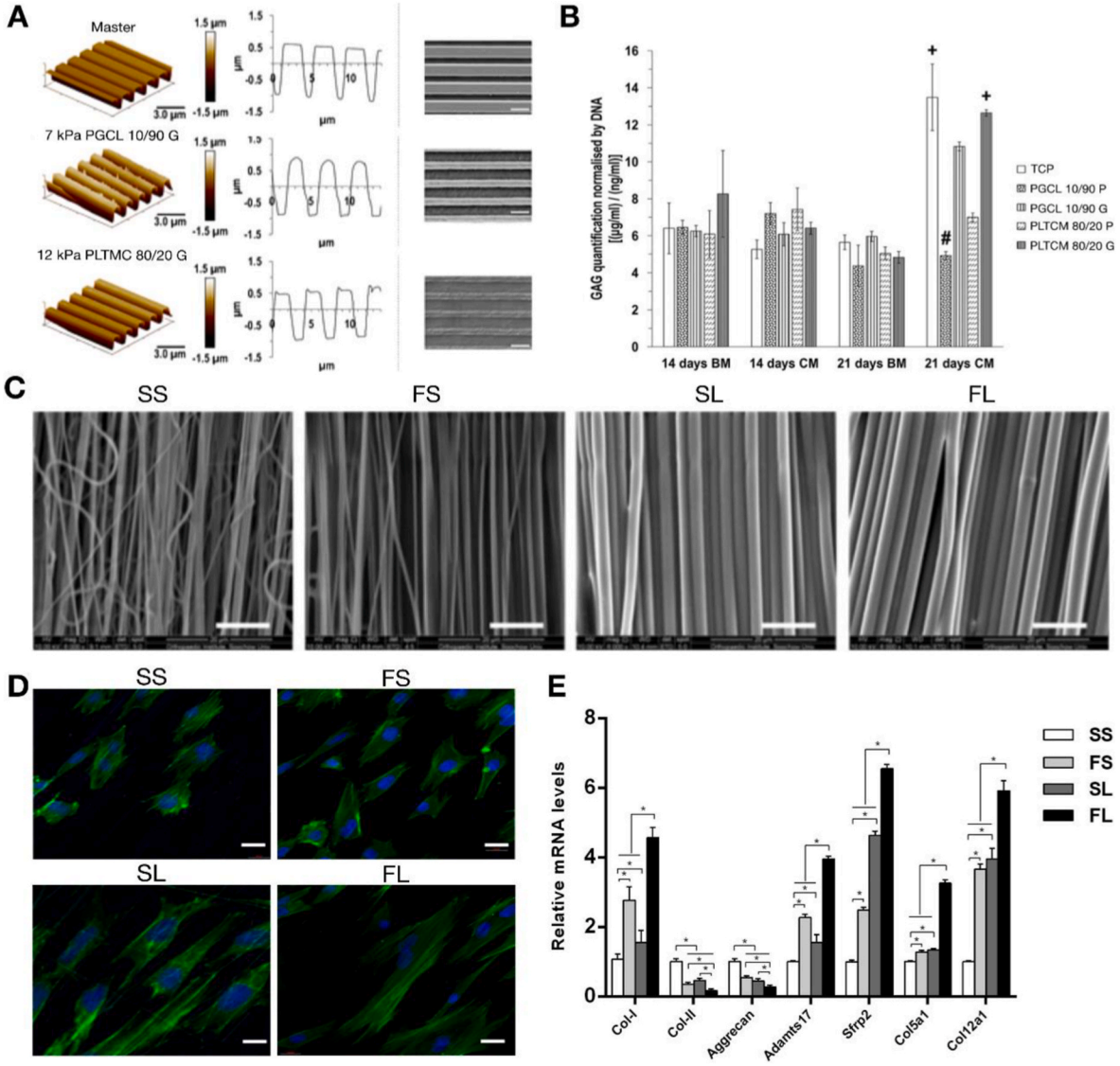


Fig. 6. Examples on anisotropic surface topographies to modulate chondrogenesis and enhance cartilage regeneration. (A) Reproducible nano- and microgrooved substrates with tunable stiffness were fabricated via nanoimprint lithography, as confirmed by AFM comparison of polymeric samples and the master mold. SEM scale bar: 4 mm. (B) High substrate rigidity allowed for an enhancement in chondrogenic differentiation when combined with nanogrooved topography. Reproduced under terms of the CC-BY license [141]. The Authors, published by John Wiley and Sons. (C) Stiffness and polymer concentration effectively allowed for the manufacturing of aligned scaffolds with controlled stiffness and fiber size, irrespective of the intrinsic mechanics in the polymers. SS: soft, small size fibers; FS: stiff, small size fibers; SL: soft, large size fibers; FL: stiff, large size fibers. (D) AFCs grown on fibrous scaffolds presented higher cytoplasmic elongation on larger, stiffer fibers (bottom row, left). (E) Stiffer, smaller scaffolds resulted in higher expression of pro-chondrogenic genes (Col-II, Aggrecan). Scale bar: 20 μm [142]. Copyright 2019 Elsevier.

wide) reduced collagen II expression and promoted fibrocartilage markers, indicating that excessive cytoskeletal polarization may compromise the hyaline chondrocyte phenotype [118,128,141].

Substrate stiffness further modulates the effects of groove topography. Microgrooved copolymers such as poly(glycolide-co-ε-caprolactone) and poly(lactide-co-trimethylene carbonate), featuring ~1 μm groove widths and elastic moduli in the soft tissue–relevant range (approximately 7-12 kPa), induced bidirectional alignment of hBMSCs and promoted GAG deposition under chondrogenic conditions. In contrast, cells cultured on planar surfaces exhibited random orientation [141]. This is consistent with previous studies reporting that micron- and submicron-scale grooves effectively guide MSC alignment and elongation compared with flat substrates, highlighting the strong influence of anisotropic topographical cues on cytoskeletal organization and cell morphology [131]. Differentiation assays performed under basal and chondrogenic conditions showed similar GAG production after 14 days. However, stiffer substrates within this range (≈12 kPa) significantly enhanced GAG deposition under chondrogenic conditions, underscoring the synergistic interplay between mechanical stiffness and topographic anisotropy in regulating cartilage matrix production [135].

In a related approach, Chou et al. created a large-scale cartilage construct by rolling microscale-grooved collagen-GAG scaffolds into cylinders, resulting in a shape that maintained the aligned architecture of the superficial zone of native cartilage [143]. The resulting constructs exhibited highly aligned collagen fibers, abundant GAG and collagen II deposition, and enhanced compressive properties (~2.1 MPa), approaching the lower range of native hyaline cartilage stiffness. Notably, narrower groove spacing (~50 μm) yielded more uniform ECM distribution, emphasizing the importance of precise microscale guidance cues in promoting chondrogenic matrix organization.

These studies show that microscale and nanoscale anisotropic topographies, especially groove-ridge architectures, play a crucial role in guiding cell alignment, elongation, and chondrogenic differentiation when their geometric parameters closely resemble native collagen

orientation. However, their effects are highly design-dependent, as excessive groove depth, suboptimal spacing, or inappropriate substrate stiffness can instead divert cells toward fibrocartilaginous or osteogenic phenotypes. Thus, careful optimization of microscale anisotropic features is essential to maintain a hyaline-like chondrocyte phenotype and harness their full potential in cartilage-focused scaffold design.

3.2.2. 3D anisotropic features: random and aligned fibrous substrates

While 2D anisotropic patterns such as grooves and ridges provide directional cues along a single plane, they do not fully recapitulate the three-dimensional complexity of native cartilage ECM. In contrast, 3D anisotropic features (e.g. fibrous scaffolds) offer a more biomimetic environment by incorporating aligned architectures in all spatial dimensions [48,144] These structures emulate the entangled collagen networks found in the superficial and transitional zones of articular cartilage, where fiber alignment and density influence both mechanical properties and cell morphology [145].

Reproducing this fibrous anisotropy has been a primary focus in scaffold design for cartilage regeneration. Electrospinning, a highly versatile technique that produces continuous submicron fibers by applying a high-voltage electric field to a polymer solution, remains a cornerstone in tissue engineering [146]. Electrospun meshes exhibit high surface area-to-volume ratios and interconnected porosity, facilitating cell attachment, nutrient diffusion, and ECM deposition. Parameters such as polymer concentration, viscosity, conductivity, flow rate, voltage, and collector geometry allow precise control over fiber diameter, alignment, and morphology [146–148].

In cartilage tissue engineering, aligned electrospun scaffolds can mimic the depth-dependent anisotropy of native ECM [149]. For example, scaffolds fabricated using poly(ether carbonate urethane)urea with tunable stiffness yielded fibers between 0.52 and 2.78 μm (Fig. 6C) [142] promoting MSC alignment along the fiber axis, particularly as stiffness fiber diameter increased as a response attributed to enhanced contact guidance and surface interaction (Fig. 6D) [150]. Similar orientation effects were previously reported in rat bone marrow MSCs on ~2 μm fibers [151]. Notably, larger-diameter fibers upregulated chondrogenic markers such as collagen II and aggrecan (Fig. 6E). Inhibition of YAP signaling, which acts as a negative regulator of chondrogenesis in MSCs, further enhanced differentiation, implicating the Hippo pathway in lineage specification [152].

Comparative studies using stable jet electrospinning have shown that fiber orientation significantly affects cell behavior. Highly aligned ultrafine fibers (~1.4 μm) were compared to randomly oriented fibers of similar diameter (~1.3 μm) [153]. Morphological differences in the resulting scaffolds (narrow, elongated pores in aligned mats versus larger, irregular pores in random ones) strongly influenced cell behavior. Adipose-derived MSCs cultured on random fibers exhibited higher SOX9 expression than those on aligned fibers, suggesting that disordered architectures may promote early chondrogenic stimulation. While aligned fibers promoted cell elongation and orientation, they were more supportive of tenogenic lineage progression. *In vivo*, these differences translated into distinct outcomes: aligned fibers supported tendon-like matrix organization, whereas random fibers led to ectopic ossification, underlining the need to fine-tune topography for specific tissue targets.

Consequently, scaffold architecture must be strategically tailored to the target tissue. Random fibers are preferable for hyaline cartilage regeneration, as they induce isotropic cytoskeletal tension and chromatin remodeling in both the cartilage-only and osteochondral regions, supporting chondrogenic markers and native cell morphology. Aligned fibers, in contrast, are better suited for fibrocartilage or tendon repair, as they engage chromatin regions that orient along the front-rear axis of the nucleus, guiding cytoskeletal organization [154].

Overall, the key topographical cues that influence cartilage regeneration are summarized in Table 1.

4. Topographical cues for subchondral bone regeneration

While surface optimization for chondral tissue engineering has received substantial attention, the subchondral bone is equally pivotal to the functional integrity of the osteochondral unit and the long-term success of implant integration. This region provides essential mechanical support and serves as a source of biochemical signals that sustain cartilage homeostasis and repair [155]. Engineering strategies that embed biomimetic topographical cues for osteogenesis are therefore critical to ensure stable cartilage regeneration and restore full joint functionality. Osteoblasts can detect and respond to nanoscale features as small as 75 nm in width and 33 nm in depth, initiating contact guidance mechanisms and cytoskeletal reorganization [156]. Recent scaffold designs increasingly exploit micro- and nanoscale surface modifications to direct cell adhesion, proliferation, and osteogenic differentiation by closely mimicking the architecture of the native bone microenvironment [157]. The following section examines scaffold-based strategies targeting the subchondral region, with emphasis on how engineered surface features can enhance osteochondral regeneration. As many of the fundamental mechanisms governing cell interactions with isotropic and anisotropic features, such as integrin clustering, focal adhesion maturation, and cytoskeletal tension, have been described in Section 3, the following section focuses specifically on osteogenic responses and bone-specific differences.

Table 1
Key outcomes of scaffolds with topographical patterns in cartilage tissue engineering.

Scale	Feature Type	Technique	Biomaterial	Dimensions	Biological effect	References
Nano	Nanopits	Polymer demixing Water-droplet templating	PCL:PMMA PLA	17-700 nm diameter; ~200 nm depth	Early SOX9 and collagen II expression; reduced cell adhesion on hydrophobic surfaces	[107,111]
Nano	Nanoislands	Polymer demixing	PCL:PMMA	90-200 nm height	Initiates collagen II expression; may impair integrin clustering if > 70 nm	[107]
Nano	Nanopillars	Electron beam lithography Plasma etching Nanoimprint lithography Injection molding	Polysterene	10 nm diameter; 62-190 nm height	Promotes SOX9, COL2A1, ACAN expression; cytoskeletal organization	[114]
Nano	Nanoholes	Thermal nanoimprinting	PCL	~225 nm diameter; 300 nm spacing	Induces polygonal morphology; supports chondrogenic differentiation	[118]
Nano	Nanotubes	Electrochemical anodization	TiO_2	50-100 nm diameter	Preserves rounded morphology; enhances GAG secretion and chondrogenic phenotype	[127]
Micro	Micropillars	Electron beam lithography and reactive ion etching + injection molding	PDMS	1-6 μm diameter; 2.4 μm height	Enhances proliferation; limited effect on chondrogenic markers	[116]
Micro	Microspherulites	Thermal annealing and quenching	PLLA	30-50 μm diameter	Sustains collagen II expression; suppresses collagen I	[130]

4.1. Isotropic patterns in subchondral bone regeneration

4.1.1. Planar isotropic features: pits, pillars, and islands

Fabrication methods like phase inversion, salt leaching, or sol-gel processes yield structures mimicking trabecular bone, with mixed micro- (1-20 μm) and macro-porosity (100-700 μm) (Fig. 7A–D) [158–160]. Micropores around 1-3 μm facilitate interlocking with the cement line matrix and support collagen deposition, while larger macropores (>250 μm) improve vascular infiltration and promote endochondral ossification [161–163]. Although cell proliferation tends to decrease on micropores smaller than 20 μm, these features still enhance early osteogenic differentiation through the phosphatidylinositol 3′-kinase/Akt signaling pathway, as indicated by increased ALP activity [164–166].

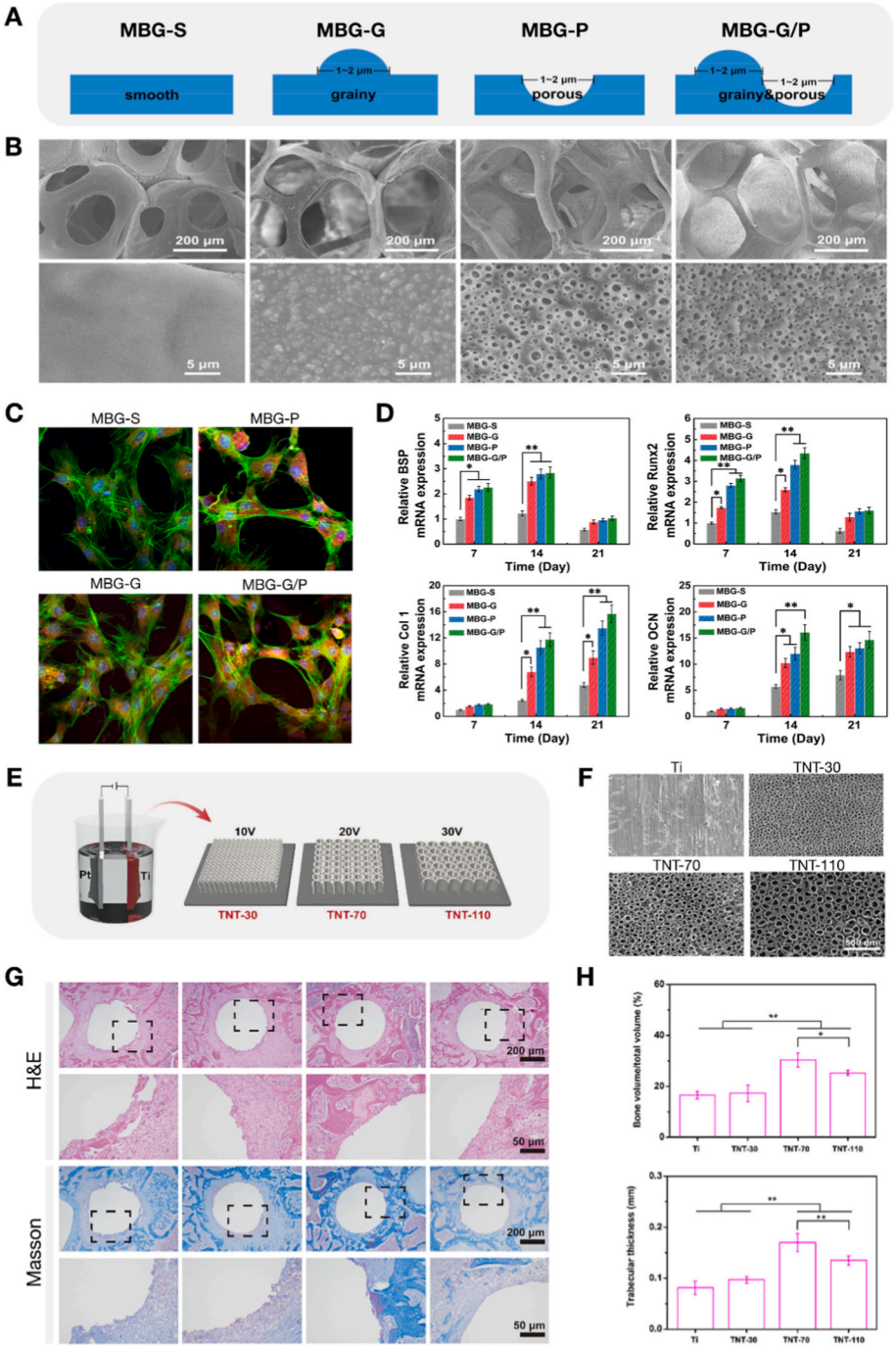


Fig. 7. Examples of tunable isotropic surface features as regulators of osteogenic differentiation and bone tissue regeneration. (A,B) Adjusting conventional surface modification techniques, such as sol-gel technique, enables the manufacturing of 2D microtextured topographies. Scale bar: 200 mm. (C) *In vitro*, rBMSCs cultured on microporous topographies developed more specific adhesion structures with dot and dash morphologies. (D) Microporous features also upregulated the expression levels of osteogenic markers (BSP, RUNX2, COL1, OCN) after 7, 14, and 21 of culture (right) [159].Copyright 2017 American Chemical Society. (E) Schematic representation of electrochemical anodization on Ti substrates. (F) Greater anodization resulted in TiO_2 nanotubes with increasing diameters of 30, 70 and 110 nm (TNT-30, TNT-70, TNT-110, respectively). Scale bar: 500 nm. (G) *In vivo,* the osteogenic response to TiO_2 nanotubes was evidenced via hematoxylin and eosin staining (first and second rows) and Masson's trichrome staining (third and fourth row); blank areas indicate to the implant position of different nanotubes, and rectangles highlight enlarged areas. (H) TiO_2 nanotubes with diameters of 70 nm demonstrated great osteoinductive potential [167]. Copyright 2023 Elsevier.

Acid-etched and sandblasted surfaces with micropits ranging from 1 to 10 μm have been shown to mimic osteoclast resorption pits, improving osteointegration by enhancing collagen type I expression and osteogenic marker levels such as ALP, osteopontin (OPN), and osteocalcin (OCN) in both human osteoblastic and murine preosteoblast cell lines [165,166]. On Ti structures, micropits between 0.5 and 20 μm significantly upregulated osteogenic genes including *RUNX2* and *OCN*, while 1 μm-diameter pits further promoted mineralization, focal adhesion maturation, and *in vivo* osseointegration, likely due to the presence of interconnected micropits, which expand the surface area available for protein adsorption, promoting osteoinduction and contributing to the bone repair process [168–171]. Lithographic techniques, on the other hand, enabled the fabrication of well-defined micropits that modulate cytoskeletal organization, with 3 μm-wide pits stimulating early focal adhesion maturation and actin activation in osteoblasts, although discontinuous arrays failed to support proper alignment [172–176].

Interestingly, the same geometric cues highlighted in Section 3.1.1 for cartilage cell lineages produce distinct outcomes in osteoblasts. For instance, pits between 5 and 14 μm (matching collagen fiber spacing in bone ECM) enhance motility and promote osteogenic cytoskeletal tension, contributing to early mineralization [173,177].

At the nanoscale, oxidative surface treatments have produced pits as small as 19-22 nm, generating highly stable and homogeneous substrates [174], or 20-22 nm, and depths of 10-20 nm [174,178,179]. Within this size range, increased β1 integrin expression triggers optimal inside-out signaling and activates FAK, promoting osteogenic proliferation and differentiation. These nanopits also support filopodia extension, an essential step in early bone tissue development at injury sites [88]. Enhanced osteogenic differentiation has also been observed on nanoporous surfaces ranging from 20 to 100 nm, with increased mineralization and OCN expression in human MSCs [180]. Moreover, these nanopits also display low inflammatory profiles by promoting M2-like macrophage polarization [178]. However, a key limitation of that study is the use of U937 macrophages, derived from lymphoma, as these are inherently more responsive towards an anti-inflammatory M2 phenotype [181], potentially underestimating the inflammatory response that would occur in primary macrophages. Nonetheless, osteoblast-specific studies confirm that excessive protrusion height (larger than 60-90 nm) reduces spreading and diminishes focal adhesion formation, correlating with reduced ALP and OCN expression [182–185].

Nanopillars and nanoislands fabricated via dry etching and phase separation (10-70 nm diameter, 8-90 nm height) similarly impact focal adhesion dynamics and osteogenic differentiation depending on size and spacing [186,187]. Optimal dimensions (~15 nm height) enhance OCN and ALP in human MSCs without stimulating osteoclastogenesis [188–190]. In contrast, taller nanopillars (>60 nm) significantly reduce focal adhesion formation and filopodia extension [191–194].

Murine osteoblasts exhibit species-specific responses, with features 100–200 nm in height inducing strong filopodia formation and marked upregulation of OCN and COL1A1 [114,195]. Even taller protrusions (800-1200 nm) have been shown to stimulate early osteogenic markers by day 7 [196,197]. On $BaTiO_3$ nanorod arrays on Ti substrates (ca. 500 nm height) regulated cell morphology and cytoskeletal organization and Piezo2/Piezo1/Ca^{2+} signaling. When coupled with a built-in electric field, these topography-mediated effects are significantly amplified, resulting in enhanced calcium-dependent signaling and improved vascularized osseointegration *in vivo* [198]. These findings indicate that protrusion height may have species-specific thresholds for osteoinduction and suggest that taller features can still be effective in certain biological contexts.

Spacing between nanoscale features further modulates cellular behavior. Nanoislands spaced 41-80 nm apart enhanced perinuclear adhesion and vinculin expression [199], while submicron pillar arrays with spacings <1 μm restricted spreading to the tops of features, reducing actin stress fiber formation and nuclear deformation [200, 201]. In contrast, nanoimprinted patterns with 1 μm or greater spacing allowed lateral substrate contact and cytoskeletal remodeling, facilitating osteogenesis [202]. Sparse patterns with 16-48 μm spacing supported FA elongation and actin anisotropy [175].

At the microscale, pillar arrays (2-4 μm diameter, 8-10 μm spacing) enhance ALP and OCN, as well as calcium deposition in MSCs [203]. Similarly, protrusions within the 1-3 μm range promote strong adhesion and spreading in osteogenic cells [204,205]. Notably, an interpillar spacing of 10 μm facilitated vascular ingrowth and bone matrix formation *in vivo* [206].

Micropillars (4-16 μm width, 4-8 μm spacing) enhance proliferation and osteogenic gene expression [207]. Heights of 1-2.5 μm support proliferation and stress fibers, although osteogenic upregulation requires inductive media [208,209]. In contrast, larger protrusions (30-80 μm high) may interfere with electric field stimulation and reduce cell viability [210,211]. Interestingly, prism-shaped features with rigid, sharp edges improved cell spreading and mechanical engagement through enhanced cytoskeletal tension [212].

Notably, human osteoblasts responded more favorably to micrometer-scale features than to nanopillars. Square prism arrays (4-16 μm width) significantly improved cell attachment and proliferation over 28 days, achieving cell densities 70-85% higher than in flat surfaces [213]. Narrower spacing (4-8 μm) strongly enhanced OCN expression and mineralization [207], consistent with prior studies showing that osteoinductive effects increase up to ~10 μm spacing, after which cytoskeletal tension decreases due to nuclear accommodation [184,214, 215].

In parallel, open-pore honeycomb architectures have emerged as promising platforms for osteochondral applications due to their mechanical robustness and vertical bone augmentation potential [216, 217]. Their high porosity and elevated rim height create a multifunctional microenvironment that supports contact guidance, nutrient diffusion, and bioactive molecule retention [218]. TiO_2 honeycombs structures (90 nm-5 μm pore spacing) modulated macrophage polarization in a size-dependent manner, with the smallest pores promoting a greater presence of podosomes via RhoA, Rac1, and CDC42 upregulation. Consistently with actin polymerization, enhanced intracellular tension combined with decreased paxillin, talin, and Src signaling promoted a shift toward the pro-regenerative M2 phenotype [219]. This immunomodulatory effect supports osteogenic differentiation in co-cultured MSCs and enhances *in vivo* osteointegration, as evidenced by increased bone-implant contact ratios and accelerated vascularization.

Grid-like microtopographies (ridge widths 2-3 μm; ridge length <3 μm) exerted cytoskeletal tension and overexpression of activated YAP, increasing ALP, OCN, and collagen I expression, with osteogenic potential declining as ridge length increases [220–222].

Together, these results show that isotropic planar topographies, ranging from nanoscale pits and pores to microscale pillars, protrusions, and honeycomb structures, can significantly boost osteogenesis by influencing focal adhesion maturation, cytoskeletal tension, integrin signaling, and mineralization in a size-dependent way.

4.1.2. 3D isotropic features: tubular and spherical structures

Tubular and spherical architectures have also been explored to enhance subchondral bone regeneration and implant integration. In particular, the subchondral bone layer plays a pivotal role in load distribution and biomechanical support for overlying cartilage, so promoting rapid and stable subchondral integration through these isotropic topographies may be critical for the long-term success of osteochondral implants. These structures provide uniform spatial cues that influence cell behavior in a non-directional manner, promoting integrin clustering, cytoskeletal organization, and nutrient exchange. Among these, nanotubular architectures fabricated via electrochemical anodization on titanium substrates have shown particular promise in improving osseointegration in mechanically demanding environments like the subchondral zone, while preserving implant integrity [215].

While TiO_2 nanotube arrays have been widely studied for cartilage-related applications, their diameter-dependent effects are particularly relevant for subchondral bone regeneration [223]. Voltage ranges between 5 and 35 V have been shown to produce well-ordered nanotube arrays with diameters spanning from 30 to 175 nm on commercial titanium surfaces (Fig. 7E–F) [167,224–226]. Among these, nanotubes with diameters between 70 and 85 nm have demonstrated superior osteogenic responses in MSCs, characterized by extensive lamellipodia and filopodia formation, as well as robust upregulation of both early (ALP) and late-stage (OPN, OCN) osteogenic markers. These outcomes are particularly advantageous for subchondral bone regeneration, where rapid matrix mineralization and strong cell-substrate adhesion are critical for the mechanical stabilization of osteochondral implants [12,39].

Consistent findings in human osteoblasts further support the efficacy of nanotubes within this diameter range, with enhanced cell spreading and mitochondrial morphological changes consistent with increased metabolic activity during osteogenic differentiation. These responses are likely linked to integrin clustering and cytoskeletal organization induced by the nanotopography. [227,228]. From a clinical perspective, achieving rapid and dense subchondral bone formation beneath cartilage-like constructs is essential to prevent fibrous tissue interposition and minimize implant micromotion. *In vivo* corroborated it, demonstrating that nanotubes within the 70-80 nm range significantly enhance bone-to-implant contact and bone volume fraction after oral implantation in mandibular bone of beagle dogs, reinforcing their potential to promote stable anchorage in the subchondral bed (Fig. 7G–H) [167,229].

In contrast, smaller-diameter nanotubes (25-35 nm), while capable of promoting modest cell elongation, exhibit limited capacity to induce osteogenic gene expression in the absence of biochemical stimulation [230,231]. This suggests that the osteoinductive effect of anodized surfaces may be influenced not only by physical dimensions but also by surface chemistry alterations introduced during anodization, and particularly in complex alloys like Ti-6Al-4V. Since Ti-based materials are widely used in osteochondral plug systems and as the subchondral phase of biphasic scaffolds, submicron-scale surface optimization holds direct implications for improving the fidelity of subchondral bone repair and long-term implant performance [39,232].

On the other end of the spectrum, larger nanotubes with diameters between 100 and 175 nm have been associated with compromised cellular outcomes. Cells cultured on these wider structures often exhibit a rounded morphology with poor spreading, correlating with diminished focal adhesion formation, reduced integrin signaling, and poor FAK phosphorylation, factors that collectively limit cytoskeletal tension and impair osteogenic differentiation [167,225]. Therefore, maintaining nanotube diameters within an optimal biological window is necessary, as both undersized and oversized features fail to support the mechano-biological cues required for functional subchondral regeneration [233,234].

Interestingly, nanotube length appears to have a limited impact on osteogenic differentiation. Studies on both titanium and zirconia substrates have shown that nanotubes ranging from 15 to 2000 nm in length support similar levels of cell adhesion, YAP/RUNX2 mechanotransduction, cytoskeletal organization, and osteogenic gene expression [235,236]. Significant biological variation has been observed primarily as a function of nanotube diameter, rather than length [233]. This length-independent behavior suggests that for subchondral applications, where vertical feature depth may vary due to implant design limitations, maintaining optimal diameter should be prioritized in scaffold design.

At the subchondral interface, curvature-dependent micro-topographies have also been investigated for their ability to regulate osteoblast mechanotransduction and matrix mineralization. While inherently isotropic, these motifs are often embedded within anisotropic frameworks such as matrix fibers, studs, or channel-like porosities, introducing direction-dependent curvature that enhances osteogenic responses [237–239]. Cells respond distinctly to concave versus convex geometries. For instance, MC3T3-E1 pre-osteoblasts on hemispherical imprints of 90 and 500 nm showed altered filopodia and migration patterns, correlated with upregulated RhoA and ROCK1 expression, whereas larger ones (1-5 μm) led to mechanical resistance and reduced motility [240]. Given that lamellipodia and filopodia measure 0.1-0.3 μm in width, features slightly below this range enable protrusions to bridge across cues with minimal hindrance [241].

Closely spaced hemispherical protrusions on titanium oxide surfaces create alternating convex and concave regions that influence osteoblast behavior. Human osteoblasts cultured on surfaces with 210-470 nm center-to-center spacing exhibited enhanced F-actin organization and longer fibronectin fibers at smaller spacings [242]. Concave architectures with negative Gaussian curvature facilitate filopodia formation by promoting inward-directed traction forces and the localization of focal adhesion components at the periphery [241]. Similarly, MSCs on small, high-curvature microspheres (0.42-0.88 μm) developed pronounced actin stress fibers and rounded focal adhesions [243]. In contrast, larger surface pores and mineralized features (10-100 μm) promoted pseudopod extension, enhanced cell spreading, and upregulated osteogenic markers such as OCN and collagen I [244,245].

Curvature effects can be amplified by mechanical cues. Convex topographies synergized with fluid shear stress to enhance Piezo1-mediated YAP nuclear translocation, leading to increased ALP, OCN, and COL-I expression [246]. PVDF films with non-porous microspheres (0.8-3 μm) also supported MSC osteogenic differentiation under inductive conditions [247]. While convex curvature can compress the nucleus and induce nucleoskeletal tension by promoting osteogenic commitment, unchanged ALP activity on both convex and concave substrates indicates that curvature alone is insufficient, highlighting the need for combined biochemical or mechanical stimulation in subchondral bone repair [238].

4.2. Anisotropic patterns in subchondral bone regeneration

Type I collagen, the primary structural protein of bone, self-assembles into staggered fibrillar arrays with a characteristic 68 nm D-periodicity, 35 nm depth, and a fine striation pattern of 5-10 nm [248]. Within this hierarchical scaffold, hydroxyapatite crystals ($\sim 50 \times 25 \times 4$ nm^3) are deposited in a highly ordered manner, reinforcing the fibrils and creating a composite mineral-collagen architecture that imparts bone its exceptional mechanical strength and resilience [249]. This intrinsic anisotropy arising from the aligned collagen fibrils and embedded mineral nanocrystals, serves as a biological blueprint for engineering surface topographies that guide cell behavior and promote osteogenic differentiation.

4.2.1. 2D anisotropic features: grooves and ridges

Biomimetic groove-ridge topographies modulate osteogenic differentiation through size-dependent effects on morphology and cytoskeletal tension. Nanogrooves promote integrin engagement and focal adhesion maturation, correlating with RhoA/ROCK activation and increased mechanotransduction [250]. At the microscale, grooves impose spatial confinement that enhances Rac/Cdc42-driven protrusive activity [251]. These scale-dependent mechanotransductive responses indicate that careful consideration of feature size is essential when designing 2D anisotropic topographies to effectively target subchondral bone.

Grooves near the size of collagen fibrils (260-410 nm) direct osteogenic alignment [252–254]. MC3T3-E1 cells consistently align on 200–400 nm grooves and maintain stable ALP activity [254]. More generally, grooves from 90 to 1000 nm induce elongation and strong focal adhesion maturation in hMSCs and osteoblasts (Fig. 8A–D) [253–257]. Elongation results from accelerated extension along the groove axis and restricted spreading perpendicular to it, a phenomenon amplified on wider nanogrooves (1200-2000 nm) [258]. However, alignment does not always equate to osteogenesis: in human periodontal

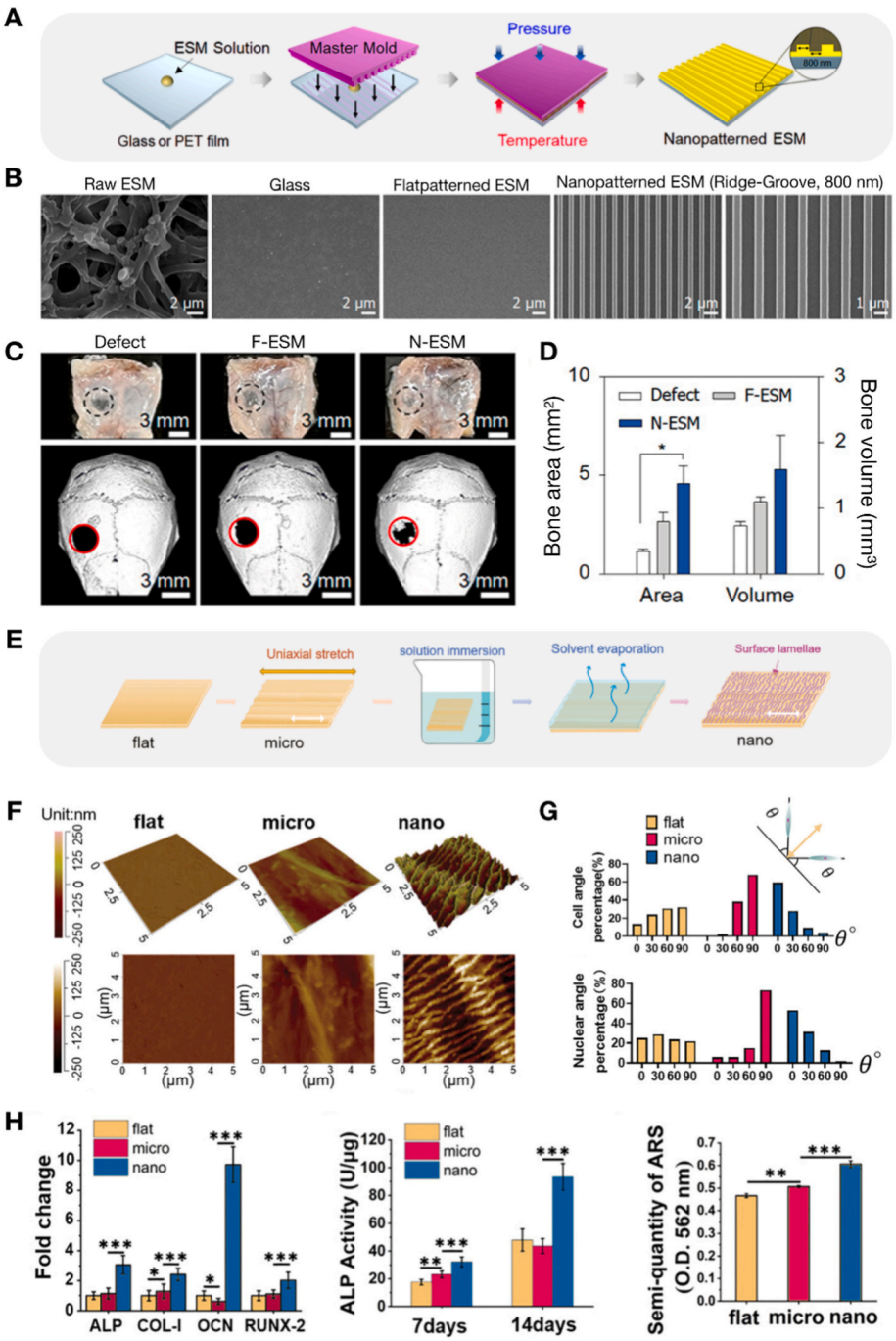


Fig. 8. Examples of tunable anisotropic surface features as regulators of osteogenic differentiation and bone tissue regeneration. (A) Schematic diagram of the eggshell membrane (ESM) repurposing to manufacture bioinspired, nanogrooved substrates. (B) SEM micrographs confirmed that the dimensions of the nano-topographical features closely matched those of the native ECM. Scale bar: 2 mm. (C) The combined effect of substrate bioactivity and topography exacerbated osseointegration on cranial bone defects after 2 weeks of patch grafting. (D) Quantitatively, significant changes in both bone volume and regenerated bone area *in vivo* were reported [257]. Copyright 2021 American Chemical Society. (E) Scheme of the fabrication of the combined orthogonal micro and nano grooves, produced by uniaxial stretching (microgrooves) and homoepitaxial crystallization (nanogrooves) (F) AFM micrographs of flat, microgrooved, and nanogrooved arrays fabricated by homoepitaxial crystallization. (G) Grooved topographies were proven to significantly modulate the polarization direction and angle of BMSCs. (H) Cellular and nuclear changes coincided with increased osteogenic gene expression [156]. Copyright 2021 John Wiley and Sons.

ligament cells, 500 nm grooves guided orientation but failed to trigger osteogenic differentiation [259], suggesting that morphological adaptation alone may be insufficient without activation of additional mechanotransduction pathways.

Depth often exerts stronger effects than width, with 400-500 nm-deep grooves promoting more mature focal adhesions and higher RUNX2 activation than shallower 100-200 nm grooves in both human osteoblasts and MSCs, even when cell spreading is oriented along the minor groove axis [260–262]. This depth dependence is more pronounced in osteoblasts than in chondrocytes, indicating a tissue-specific mechanosensitivity to vertical confinement.

Because osteochondral repair involves cells spanning several micrometers interacting with an ECM that contains fibers with tens of micrometers in length, responses to anisotropic features emerge from

multi-scale interactions [263]. Combining micro- and nanogrooves coordinates polarization via distinct pathways (Fig. 8E–F). Microgrooves suppress actomyosin tension, whereas nanogrooves enhance adhesion maturation and F-actin alignment via RhoA/ROCK signaling (Fig. 8G–H) [156].

At the microscale, groove-to-ridge ratios strongly regulate osteogenesis. Features 5-15 μm wide enhance elongation, whereas widths >15-20 μm reduce directional sensing and lower intracellular tension [264–266]. Conversely, very narrow grooves (<5 μm) confine adhesions to ridge tops, limiting effective force transmission and attenuating osteogenic commitment. [267,268]. This mechanical constraint correlates with reduced myosin contractility and lower osteogenic differentiation, as shown by decreased ALP activity compared with flat controls. Interestingly, ca. 1 μm grooves stabilize podosomes by reducing myosin II-based contractility [269], a hallmark of osteoclastogenesis that facilitates bone resorption [270]. Larger features reveal additional geometry-dependent behaviors. For example, while 15-20 μm grooves maintain alignment, widths exceeding this threshold shift focal adhesions toward groove centers, decreasing cytoskeletal tension and nuclear deformation [271–273]. MC3T3-E1 cells also show reduced adhesion on features >50 μm, reflecting curvature-dependent mechanical sensing [274,275]. Smaller radii of curvature amplify α-smooth muscle actin incorporation and vinculin engagement through RhoA activation, whereas larger radii fail to concentrate tension and instead promote cytoskeletal relaxation [276]. Interestingly, macroscale grooves ca. 200 μm still stimulate YAP nuclear translocation, likely because they mimic trabecular spacing in cancellous bone. In murine models, YAP accumulation and its coupling to the YAP-autophagy signaling axis drive strong osteogenesis, suggesting that certain large-scale geometries can bypass conventional alignment-driven pathways by activating mechanosensory programs specific to the subchondral bone environment [276,277].

The reported results indicate that osteogenic effects depend less on width alone and more on the groove-to-ridge ratio, which governs effective mechanotransduction signaling. Excessively wide ridges reduce the frequency of cell-topography interactions, thereby diminishing hMSC alignment and lineage commitment [256]. For example, substrates with 2 μm grooves and 6 μm ridges (1:3 ratio) induced higher osteogenic mRNA expression than those with 4 μm grooves and 11 μm ridges [278]. Similarly, when groove width was held constant (200 μm), increasing ridge width from 25 to 200 μm progressively reduced topographical guidance and osteogenic activity to levels comparable to flat controls [279,280]. Maintaining a high interfacial density therefore appears essential for generating the dense subchondral bone layer required for stable cartilage support.

Beyond osteogenesis, groove geometry also influences immune cell behavior. Macrophages cultured on 2D grooves fabricated on tissue culture substrates using ECM stamps of 20 μm and 50 μm widths showed markedly different polarization profiles [90]. Enhanced cell elongation on the narrower grooves increased expression of arginase and CD206, markers of the anti-inflammatory M2 phenotype, under both basal and IL-4/IL-13-stimulated conditions. While elongation did not significantly alter iNOS expression in LPS/IFNγ-stimulated macrophages (M1-like phenotype), secretion of pro-inflammatory cytokines was reduced on the smaller grooves. These results suggest that geometric control of macrophage shape can bias polarization toward an anti-inflammatory state, likely through actin-dependent mechanotransductive pathways that regulate cytoskeletal tension and downstream signaling.

As discussed earlier, substrate stiffness further modulates these topographical effects and can partially compensate for non-optimal groove geometries. On microgrooved surfaces, increasing stiffness to approximately 12-46.7 kPa enhances cell alignment, reduces cellular circularity, and modestly promotes osteogenic differentiation, even when groove–ridge dimensions are not optimal [141,281]. Therefore, integrating optimized groove-ridge ratios with stiffness cues tuned for subchondral bone could yield a more robust and anchored mineralized layer, improving long-term cartilage regeneration outcomes.

4.2.2. 3D anisotropic features: random and aligned fibrous substrates

Electrospun fibrous scaffolds offer a versatile platform for engineering anisotropic environments that mimic the structural organization of subchondral bone. Through careful control of electrospinning parameters, such as mandrel rotation speed, polymer solution viscosity, and jet dynamics (e.g., whipping and bending instabilities), fiber alignment and diameter can be tuned to influence cell behavior [282,283]. Mechanistically, fibrous topographies potentiate integrin-mediated signaling, activating ILK, RhoA, and YAP/TAZ pathways that promote osteogenic commitment [284,285].

Several studies have revealed that the osteoinductive potential of aligned nanofibers is highly context-dependent, influenced by cell type, scaffold composition, and environmental stimuli [286–288]. For instance, aligned PLLA nanofibers have been shown to enhance osteogenesis of adipose-derived MSCs by upregulating osteogenic genes. Similarly, hydrostatic pressure applied to aligned and random PCL fibers synergistically promoted osteogenic differentiation, although random fibers under mechanical stimulation led to stronger mineralization and expression of ALP, RUNX2, and collagen I [286,288]. In contrast, aligned fibers have also been reported to favor tenogenic differentiation over osteogenesis. Recent studies reported that aligned nanofibrous scaffolds suppressed osteogenic markers and promoted tendon-like tissue formation, while randomly oriented fibers induced chondrogenesis and ectopic bone formation *in vivo* [289].

Beyond alignment, fiber diameter and architecture critically influence osteoinductive outcomes. Fibers <1 μm in diameter drive nuclear and cytoplasmic alignment with osteoinductive potential [285], but dense submicron mats can restrict migration, ECM deposition, and angiogenesis [290,291]. To overcome this limitation, hierarchical mats combining ~2.93 μm microfibers branching into 212 nm nanofibers have been developed, mimicking osteoclast-resorbed bone surfaces and supporting trabecular bone formation over time [292]. Similarly, embedding ~6.5 μm beads within nanofibers created biomimetic voids that increased bone sialoprotein expression, showing the size-dependent osteoinductive capacity of fibrous scaffolds [293].

Fiber diameter also governs pore size, a key factor for bone-like mineralization. Microfibers (0.14-2.1 μm) yield larger pores and greater cell infiltration than nanofibers (200-400 nm) [294,295]. Random and aligned microfibers (18-20 μm) produce pores (~200 μm) within optimal ranges for osteoblast proliferation (200-350 μm) and post-implantation mineralization (100-200 μm) [102,296], supporting subchondral bone volume restoration [297]. In both nano- and microfibrous systems, elongation and alignment along fibers occur in early stages of culture. This alignment transmits cytoskeletal forces to the nuclear envelope via LAMIN A/C protein, suggesting that fiber orientation acts as an epigenetic cue for osteogenesis [296,297].

While random fibers can enhance proliferation, chromatin accessibility, and metabolic activity [288,298], their lower tensile strength limits their ability to sustain osteogenesis, particularly in load-bearing subchondral regions. Aligned mats provide higher stiffness and tensile strength, better approximating native bone mechanics and supporting calcium deposition and OCN expression [287]. ALP activity appears highest on ~400 nm fibers (both aligned and random) and is unaffected in microfibrous scaffolds [294,295], possibly due to the transient nature of ALP and insufficient mechanical stimulation from densely packed nanofibers. Introducing secondary anisotropic cues, such as grid patterns constraining random fibers, boosted ALP activity 2.7-fold compared to unpatterned controls [299], indicating that combined architectural hierarchies may be most effective for directing osteogenic differentiation in osteochondral repair.

Gridded PCL scaffold meshes produced by melt-electrowritting of pore sizes ranging from 100 μm down to 40 μm have demonstrated a pore size dependent approach on macrophage elongation in 3D [300]. Smaller pore meshes direct the spontaneous differentiation of

monocyte-derived macrophages while larger pore sizes induced enhanced phagocytic activity indicative of an M1 phenotype.

Overall, the key topographical cues that influence subchondral bone regeneration are summarized in Table 2.

5. Topographical gradients design for osteochondral regeneration

Building on the cartilage- and bone-specific mechanisms discussed in Sections 3 and 4, no single topographical cue can recapitulate the hierarchical, anisotropic nature of the osteochondral unit. Instead, functional gradients that combine chondrogenic isotropic/anisotropic features with osteoinductive cues for the subchondral bed constitute a pragmatic path toward biomimetic design. The rapidly expanding evidence that surface topography modulates cell morphology, mechanotransduction, and lineage commitment across both tissues motivates scaffold architectures that intentionally integrate multiple cues within a continuous design framework [302,303]. In line with this, the field is moving beyond uniform or single-scale surfaces toward multi-topographical models that guide spatially distinct responses, either reproducing the cartilage-to-bone transition or mimicking the depth-dependent zonal organization of native articular cartilage.

Strategies that reproduce the anisotropic microenvironment of cartilage frequently combine gradients in surface cavities with directionally aligned substrates to impose zone-specific guidance [100]. Scalable methods, such as solvent casting, particle leaching, and phase separation, enable pit arrays whose dimensions can be tuned by porogen size and concentration or polymer viscosity, allowing layer-by-layer control of surface features [304]. Building on this principle, leaching gelatin microspheres of different diameters from PLCL matrices yields multilayered scaffolds with surface pits of 13 or 47 μm, while layer orientation and microsphere loading further tailor the microenvironment of the superficial, middle, and deep zones (Fig. 9A) [305,306]. Across these platforms, larger micropits that approximate native lacuna dimensions tend to enhance cartilage-specific responses relative to smaller features. Notably, 47 μm pits with shallow deposition angles reduced GAG deposition yet produced the highest cell density, indicating improved cell accommodation with moderated matrix synthesis (Fig. 9B–C); in contrast, more open architectures with lower microsphere loading diminished initial hMSC attachment but improved mechanical performance, underscoring the need to balance biological and structural outcomes when designing zonal topographies.

The interplay between surface micropits and intrinsic scaffold porosity can confine cells within lacuna-like microenvironments in which cytoskeletal tension and localized hypoxia drive zone-specific matrix production [308]. This response mirrors the native depth-dependent composition of cartilage and highlights microscale cavities as chondrogenic niches, an aspect often overshadowed by nanoscale features. As single-topography exemplars, silk fibroin scaffolds produced by directional freezing generate horizontal, random, and vertical architectures with 45-80 μm pits [309]. *In vivo*, vertically oriented patterns favored both cartilage and subchondral compartments and supported appropriate calcification in deeper regions. Because these cues were tested independently rather than within an integrated osteochondral gradient, their ability to support mature hyaline cartilage formation was limited, emphasizing the necessity of reproducing the native zonal hierarchy instead of relying on a single architectural cue.

To directly encode zonal organization, prior work has assembled anisotropic nano-to microscale topographies rather than relying solely on microscale pits. For example, aligned and random PLA electrospun meshes (518 and 486 nm fibers, respectively) have been layer-by-layer integrated atop a hyaluronic acid hydrogel containing 500 μm vertical microgrooves to reproduce the superficial-middle-deep anisotropy (Fig. 9D) [149]. In this construct, bovine chondrocytes exhibited a zonal deposition pattern, with collagen II and aggrecan enriched in middle/deep regions and inversely correlated with collagen I (Fig. 9E). These findings corroborate the chondrogenic efficacy of nanofibrous topographies, where random fibers induce isotropic cytoskeletal tension and prime early chondrogenesis via chromatin remodeling, whereas wider microgrooves favor cartilage ECM deposition [141,142,151]. Together, they illustrate how multi-scale anisotropy can be layered to emulate depth-dependent guidance cues.

Transitional biomimicry from cartilage to bone can also be achieved by coupling topographical and biochemical cues within a single

Table 2
Key outcomes of scaffolds with topographical patterns in bone tissue engineering.

Scale	Feature Type	Technique	Biomaterial	Dimensions	Biological effect	Reference
Nano	Nanopits Nanopores	Chemical treatment, Electron beam, Lithography	Ti PMMA	~20-22 nm diameter; ~10-20 nm depth	Reduced inflammation, enhanced filopodia anchoring	[179,180]
Nano	Nanopillars Nanoislands	Polymer etching, Block copolymerization, Plasma etching, Anodization & deposition, Injection/ Nanoimprint molding, Electron beam lithography	Polysterene:poly(4-bromostyrene), Poly(styrene-block-poly-2-vinylpyridine), Silicon, SiO_2, Ti, Ti:HA,TiO_2, PMMA	70-200 nm diameter; height ≤50 nm; interspacing ~600 nm-1 μm	Enhances cell adhesion, focal adhesion maturation, and upregulates ALP and OCN expression	[108,109,114, 185–187,190, 193–195,301]
Micro	Micropits Micropores	Phase inversion, Sol-gel/ multitemplate, Replication & coating, Acid/grit treatments, Lithography & etching, Monomer casting, Polymer demixing	PCL:HA, CaP:HA, Ti, Polysterene	1-20 μm, critical size ~1-3 μm	Enhances osteoblast differentiation and increases ALP activity	[158–160,162, 163,165,166, 171,175,176]
Micro	Microgrooves	Photolithography, Mold replication, Laser engraving, Thermal/acid etching, Monomer casting, Electron beam/nanoimprint lithography	PCL, Ti, PDMS, PLLA-PCL, Poly(3-hydroxybutyrate-co-3-hydroxyhexanoate), β-tricalcium phosphate	Width: 1-4 μm; Depth: ≥317 nm; Spacing: 2-10 μm	Promotes cell alignment, elongation, and osteogenic gene expression via contact guidance	[145,243,244, 247,249–251, 253,255,256] [156,260,261, 264,267–269, 271,273,274]
Micro	Micropillars Grids	Injection/hydrothermal molding, Thermal dewetting, Soft lithography, Photolithography & plasma treatment, Nanoimprint, Femtosecond irradiation	Polysterene, Ti, Nanoporous alumina, PMMA, PDMS	Features: 1-4 μm; spacing ~8-10 μm (also ranges up to 4-16 μm)	Increases osteogenic marker expression and cell proliferation	[190–196,207, 208] [203–209,220, 221]
Macro	Ridges	Photolithography & hot embossing, Soft photolithography, Electron beam lithography, Nanoimprint	Polysterene, PCL, Ti, Polyacrylamide, Poly(l-lactide-co-trimethylene carbonate)	Ridges: 2-3 μm wide; ridge length ≤3 μm; optimal groove:ridge ratio ~1:3	Elevates ALP, OCN, and collagen I expression; overly wide ridges diminish osteogenic effect	[131, 258–261] [141, 278–281]

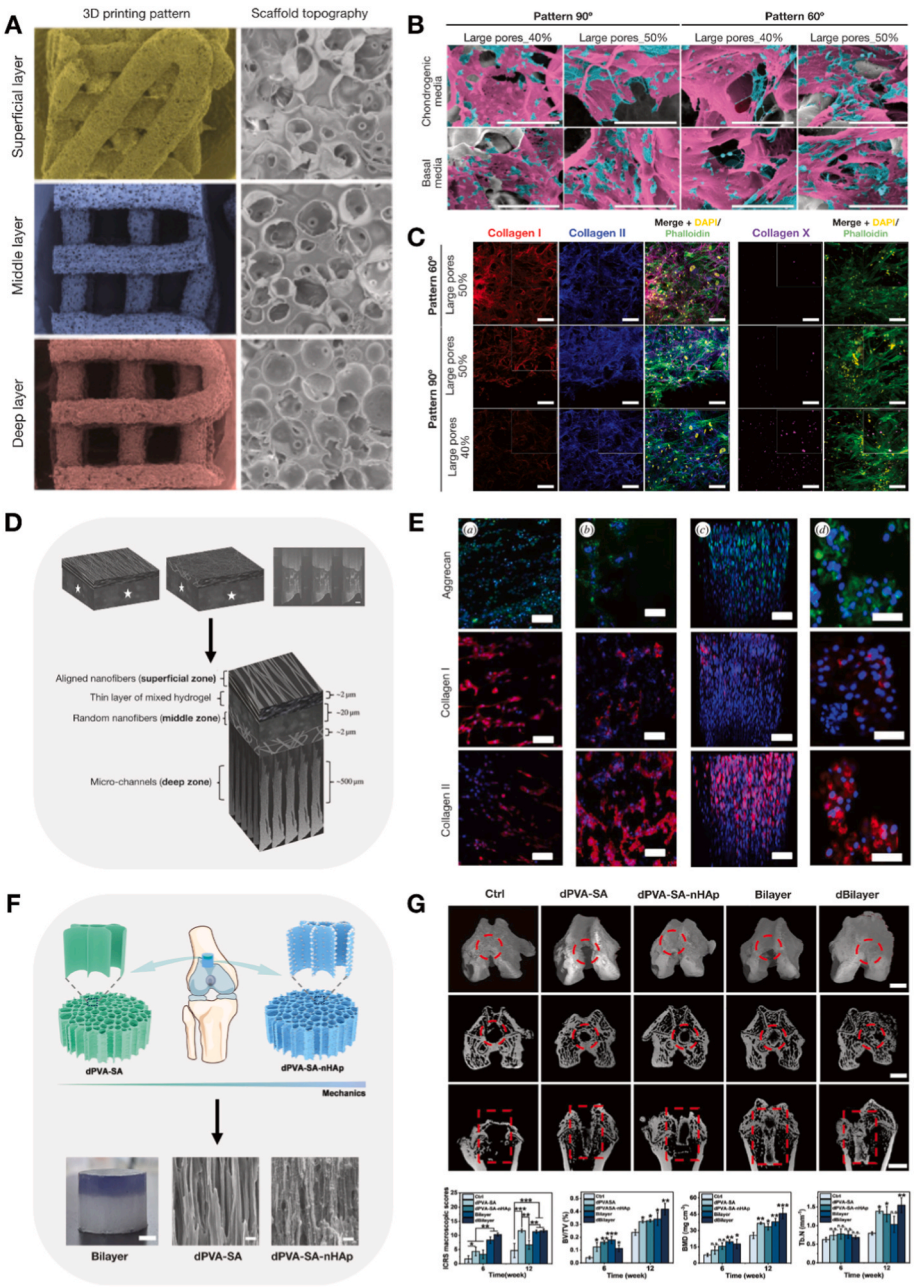


Fig. 9. Examples of complex topographical scaffolds regulating articular cartilage and subchondral bone regeneration in osteochondral tissue engineering. (A) Microsphere-leached PLCL matrices enabled the 3D-printing of multilayer scaffolds with gradient porosity and isotropic microscale features. (B) Micropit topographies with controlled porosity promoting fibrillar-like ECM deposition (blue) and defined cell morphology (pink). Scale bar: 30 μm. (C) Fluorescence imaging of hMSCs cultured for 28 days on dual-porosity multilayer scaffolds showing depth-dependent collagen expression: collagen II (red) maintained throughout the construct, collagen I decreasing with depth, and collagen X localized to the deepest layer. Scale bar: 100 μm [305]. Copyright 2025 Elsevier. (D) Schematic side view of zonal scaffolds shown isolated (top) and assembled (bottom). (E) Immunostaining of bovine chondrocytes demonstrated increased collagen II and aggrecan in middle and deep zones, with depth-dependent reduction of collagen I and progressive aggrecan accumulation. Scale bar: 100 μm [149]. Copyright 2018 The Royal Society. (F) Bilayer hydrogel scaffold with directional microgrooves fabricated via ice-templating and double-crosslinking of PVA:SA; SEM images show vertically aligned channels in PVA-SA and PVA-SA-nHAp constructs. Scale bar: 10 μm. (G) *In vivo* evaluation showed enhanced osteochondral integration driven by the mechanical gradient and vertical microgrooves, evidenced by micro-CT reconstructions after 12 weeks and quantitative repair assessment using International Cartilage Repair Society (ICRS) scoring and bone morphometric parameters (BMD, BV/TV, Tb.N) [307] Copyright 2025 John Wiley & Sons, Inc.

platform. Vertical microgrooves ~13 μm wide, introduced by direct templating in PVA:sodium alginate hydrogels containing dispersed nanohydroxyapatite, effectively recapitulate both compartments in one fabrication step (Fig. 9F) [307]. The anisotropic grooves promote cell orientation and cytoskeletal alignment and are associated with COL2 upregulation, consistent with cartilage constructs using similar grooved patterns. Although the nanohydroxyapatite contributes osteoinductive signals, the grooves themselves appear to be the primary topographical driver of alignment. In parallel, directional microgrooves elevate COL1, OPN, and OCN, underscoring their osteogenic potential, which was further translated to greater tissue integration *in vivo* (Fig. 9G). Collectively, these data show that an appropriately engineered single surface topography, tuned in composition and mechanics, can simultaneously enhance cartilage and bone regeneration.

6. Conclusions and future perspectives

The regeneration of osteochondral tissue remains challenging as cartilage and subchondral bone present distinct yet interdependent biological and mechanical environments. As discussed previously and summarized in Table 3, topography is a central design parameter that shapes cell behavior in a zone-specific manner. In cartilage-like regions, isotropic features such as nanopits (400-700 nm width, ~200 nm depth), nanoislands (90-200 nm height), and spherical microdomains (30-50 μm diameter) help maintain a rounded morphology and support hyaline-associated markers (e.g., SOX9, collagen II). In contrast, anisotropic cues, namely grooves (50-1000 nm width) and aligned fibers (0.5-2.8 μm diameter), promote elongation and alignment that can bias toward fibrocartilage or tenogenic phenotypes if not balanced by appropriate mechanics and biochemical signals. In subchondral bone, nanoscale features that foster integrin clustering and focal adhesion maturation, such as nanopits (~20-22 nm), nanopillars (70-200 nm diameter, ≤50 nm height), and nanotubes (optimal at 70-85 nm diameter), enhance osteogenesis, while microscale grooves and macroporosity support vascular infiltration, mineralization, and mechanical stability. These effects, however, are also highly reliant on substrate stiffness and the incorporation of biochemical cues.

Dimensionality is also a crucial factor for osteochondral scaffold design. 2D features (e.g., grooves, ridges) can strongly influence adhesion patterns and cytoskeletal organization, yet they are often insufficient to support osseointegration over time. 3D architectures, including nanofibrous matrices and lattice-based constructs, provide the complex cell-cell/cell-matrix signaling and interconnectivity required for cell infiltration, nutrient diffusion, and matrix deposition. However, they typically demand more time-consuming fabrication methods. A forward-looking strategy for osteochondral scaffold design is to integrate both 2D and 3D topographical cues within a single construct and to transition between these features gradually along its depth, thereby satisfying the distinct biological and mechanical demands of each region within the osteochondral unit. Moreover, topography alters local force transmission and focal adhesion patterns, changing the perceived stiffness even when the bulk material remains unchanged. Similarly, topography influences fluid flow across or through a scaffold, adjusting local shear stresses and strain distributions during mechanical loading. Additionally, topography can affect endogenous electrical signals by modifying charge distribution or piezoelectric responses during deformation. Therefore, topography ultimately modulates how strongly cells respond to stiffness, shear, and electrical cues, and should be considered when designing osteochondral scaffolds.

A persistent challenge is extrapolating monoculture results to environments where direct cell-cell contact and paracrine signaling govern outcomes. Because scaffold architecture influences diffusion pathways, pore connectivity, and spatial proximity between populations, single-lineage readouts provide only a partial view of biological impact. To bridge this gap, computational approaches present a promising way to connect single-cell observations with tissue-level behavior. Models that describe how individual cell types detect geometry can be integrated with simulations of soluble factor transport within complex microarchitectures to forecast how cells influence each other over time. Embedding such predictive tools into the design workflow and validating them with targeted co-culture experiments should improve the selection of architectures that stabilize a chondrogenic phenotype in the upper zones while promoting mineralized matrix deposition and vascular ingrowth in the subchondral region.

Translating these insights into clinically meaningful constructs requires manufacturing strategies that balance architectural precision with scalability and reproducibility. Techniques such as electrospinning, melt electrowriting, and high-resolution additive manufacturing can deliver fine control over feature size and orientation. Still, they must be optimized to sustain throughput, reduce batch variability, and ensure reliable integration between layers and zones. Addressing persistent issues, including mechanical instability, interfacial delamination, and uneven maturation, will depend on co-optimizing geometry and mechanics from the outset and benchmarking performance in standardized test systems that more closely mimic *in vivo* loading and transport conditions.

Leveraging biophysical topography instead of exogenous drugs or growth factors also offers a safer translational and regulatory path. Yet, regulatory agencies will still require clear evidence that topographical features can be manufactured with consistent fidelity and retain their function under physiological stress. Meeting these expectations

Table 3
Optimal parameter ranges for effective osteochondral tissue engineering.

Tissue	Feature type	Optimal range	Suboptimal range	Mechanism	Optimal biological outcome	Suboptimal biological outcome
Cartilage	Nanopits	Diameter: 200-400 nm	Diameter >700 nm	Integrin spacing Focal adhesion maturation	Hyaline phenotype maintenance (↑ SOX9, COL2)	Reduced integrin clustering
	Nanoislands	~100 nm	>70 nm	Adhesion destabilization	Early chondrogenic priming (↑COL2)	Impaired integrin clustering
	Nanopillars	Height: 150-200 nm	Height <75 nm	Vertical mechanosensing Nuclear sparing	Preserves rounded morphology (↑SOX9 and ACAN)	Behaves like a flat control
	Grooves	Width: ~100 nm Moderate depth	Width ≥317 nm	Cytoskeletal polarization	Cartilage alignment without de-differentiation	Induction of RUNX2 drift
	Spherical domains	Lacunae-mimetic (~30-50 μm)	Non-spherical, amorphous	Low nuclear strain	Hyaline ECM deposition (↑SOX9, COL2, and AGG)	Loss of hyalin phenotype
Bone	Nanopits	Depth: 20-50 nm	<20 nm	β1-integrin clustering FAK pathway	Osteogenic differentiation (↑OPN and OCN)	Impaired mature adhesion formation
	Nanopillars	Height: 15-50 nm Interspacing >1 μm	≥90 nm Interspacing <1 μm	Traction perinuclear tension YAP translocation	Mineralization onset (↑ALP and OCN)	Reduced cell spreading and nuclear deformation
	Nanotubes	Diameter: 70-85 nm	<30 or >100 nm	RhoA, YAP/TAZ	Strong osseointegration (↑RUNX2 and OCN)	Poor osteoinductive effect
	Microgrooves	Width: 5-15 μm	<5 μm; >20 μm	FAK/ROCK tension Nuclear strain	Osteogenic polarization (↑ALP)	Shallow grooves confine the cytoplasm to the ridge top Wider grooves reduce cytoskeletal tension
	Micropores	Diameter: 1-20 μm; >250 μm	<1 μm; non-interconnected, large lattice/discontinuous	Curvature-dependent nuclear strain	Vascularized bone ingrowth (↑ALP)	Induces excessive curvature Large lattices behave like flat controls

demands physiologically relevant bench testing, long-term *in vivo* validation in large-animal models, and strict alignment of fabrication workflows with Good Manufacturing Practice standards. These include defined process controls, post-processing, and rigorous dimensional characterization, to ensure that the final construct reliably preserves its intended architectural and functional attributes.

In the near term, progress will rely on multiscale scaffolds with controlled mechanical gradients, integrated 2D/3D features, and reproducible zonal architectures. Looking further ahead, the development of stimuli-responsive or temporally evolving topographies may provide additional flexibility by presenting stage-appropriate cues during tissue maturation. Although these adaptive systems remain technically demanding, they represent an exciting frontier in the field. Ultimately, the convergence of advanced biofabrication, mechanobiology, and immunomodulatory design with biomimetic, multiscale architectures brings the field closer to achieving long-term osteochondral repair, combining stable mineralized subchondral engraftment with durable cartilage integrity while advancing toward clinically feasible, industrially scalable solutions.

CRediT authorship contribution statement

Lucia Aboal-Castro: Writing – original draft, Investigation, Conceptualization. **Vasiliki K. Kolliopoulos:** Writing – original draft, Formal analysis, Conceptualization. **Carmen Alvarez-Lorenzo:** Writing – review & editing, Formal analysis, Conceptualization. **Antonios G. Mikos:** Writing – review & editing, Formal analysis, Conceptualization. **Luis Diaz-Gomez:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization.

Ethics approval and consent to participate

This study did not involve human subjects, patients, or animals. Therefore, ethical approval or consent to participate was not necessary.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in this article.

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